

Week 6

Microbiome Bioinformatics – From Community Data to Scientific Discovery

Microbiome Assays, 16S/Shotgun Metagenomics,
Ecological Interpretation, and AI-Assisted Discovery

Bioinformatics: From Multi-Omics Data to Discovery
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Two 40-Minute Sessions | Fall 2026

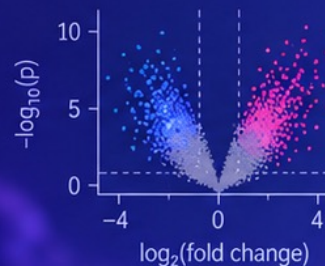
16S rRNA

Shotgun
metagenomics

Taxonomic composition



Differential abundance



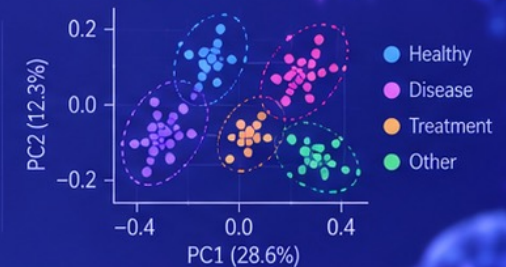
Abundance heatmap



AI-assisted
interpretation



PCoA

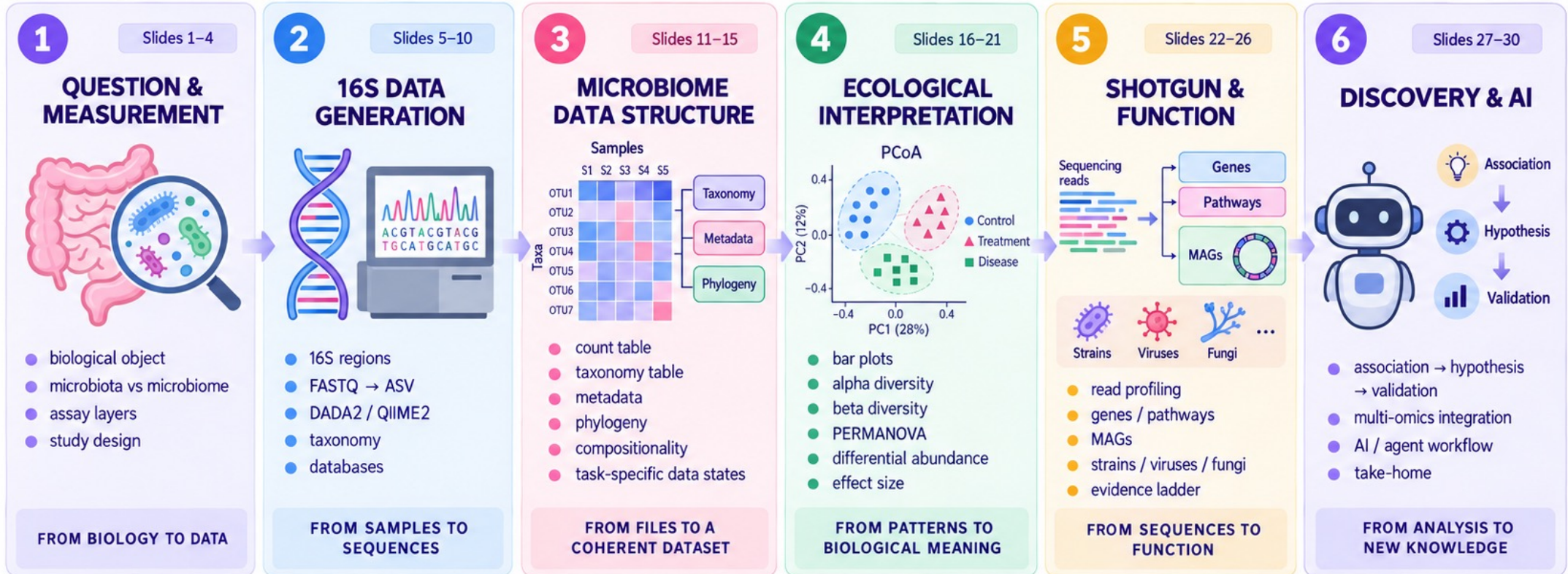


Week 6 Course Architecture — From Microbiome Data to Scientific Discovery

30 slides organized around biological questions, data structures, interpretation, and evidence.

Data • Microbes • Ideas • Healthier Lives

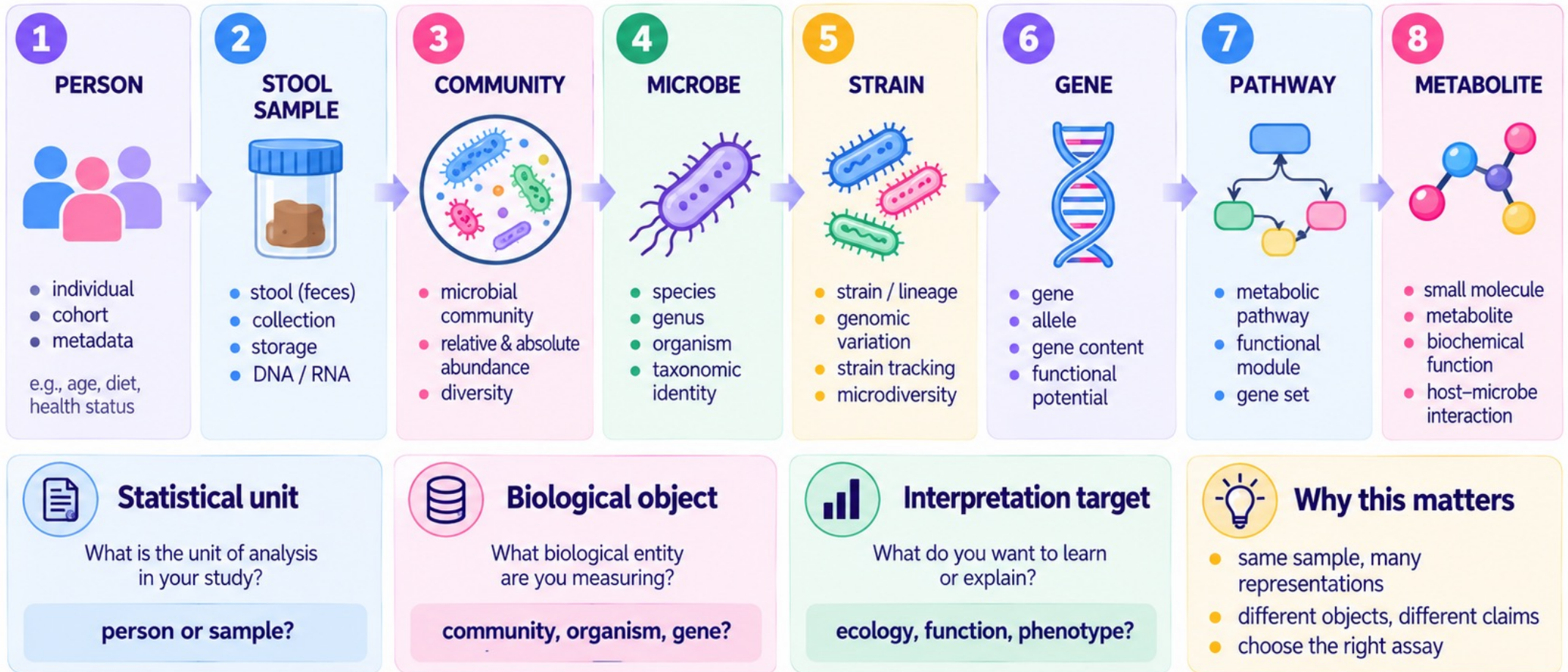
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|
Question first. Assay second. Interpretation last.

What Exactly Is the Biological Object?

Microbiome studies compare people, samples, communities, and molecular layers.



Define the biological object **before** choosing the assay or model.

Microbiomes span the gut, body, and environment

A microbiome is a community of bacteria, archaea, fungi, viruses, and other mobile genetic elements living in a defined habitat.

Major habitats

Human-associated habitats

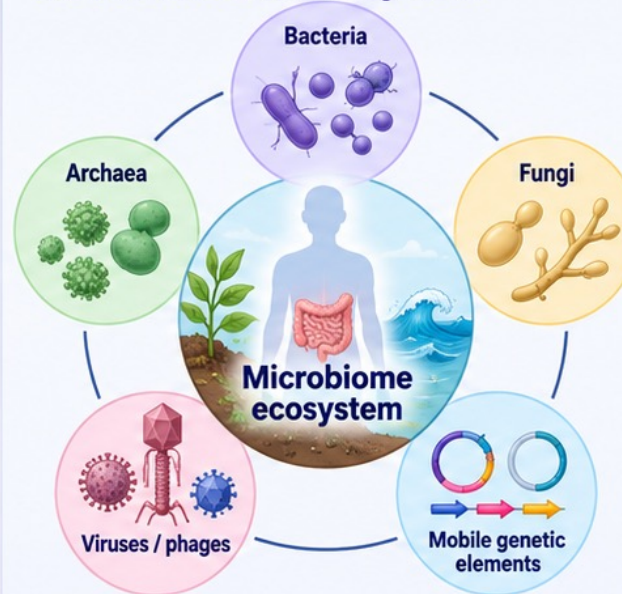


Environmental habitats



- Each habitat has distinct nutrients, oxygen, pH, and host or ecological constraints.
- The habitat shapes which microbes can survive and interact.

The microbiome ecosystem



Different members interact through competition, cooperation, predation, and metabolite exchange.

What each group contributes

Bacteria	Archaea	Fungi	Viruses / phages	Mobile elements
 • Abundant community members • Metabolism • Fermentation	 • Methanogenesis and specialized metabolism	 • Eukaryotic microbes • Colonization • Host interaction	 • Infect microbes or host cells • Shape community structure	 • Plasmids and transposons • Horizontal gene transfer

Why this matters for microbiome research



- Community membership is not the same as activity.
- Different assays capture different layers of evidence.
- Microbiome should not silently mean only bacteria.

Common measurement strategies

Different assays are needed for different questions.



16S rRNA / amplicon sequencing
Bacteria and archaea
(community composition)



ITS sequencing
Fungi
(community composition)



Shotgun metagenomics
All microbial kingdoms
(functional potential, mobile elements)



Metatranscriptomics
Active genes
(gene expression)



Metabolomics
Microbial and host
metabolites (functions)



Host phenotyping
Host or ecosystem
traits and health



Take-home message: A microbiome is a multi-kingdom ecosystem whose members and functions depend on the habitat being studied.

One Sample Can Generate Many Different Data Types

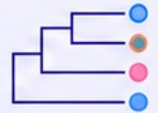
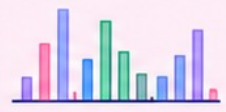
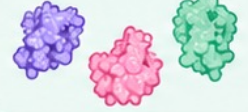
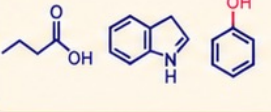
The same specimen can produce different data matrices and support different biological claims across bacteria, archaea, fungi, viruses, and other microbes.



The assays and markers determine the **data structure** and the **microbial groups** you can study.

Different Assays Measure Different Biological Layers

Taxonomy, genetic potential, activity, protein output, and chemistry are related but not equivalent.

Biological layer	Assay (method)	Question answered	Output examples
1 Community membership 	 16S / marker-gene sequencing	Who is detected?	 ASVs, taxonomy, phylogeny
2 Genetic potential 	 Shotgun metagenomics	Which genes and pathways exist?	 species, strains, genes, MAGs
3 Active transcription 	 Metatranscriptomics	Which microbial genes are expressed?	 RNA abundance, pathway activity
4 Protein output 	 Metaproteomics	Which proteins are detected?	 enzymes, expressed functions
5 Chemical output 	 Metabolomics	Which small molecules accumulate?	 SCFAs, bile acids, indoles, xenobiotics



Interpretation rule: Presence of DNA does not by itself demonstrate expression, protein activity, metabolite production, or causality.

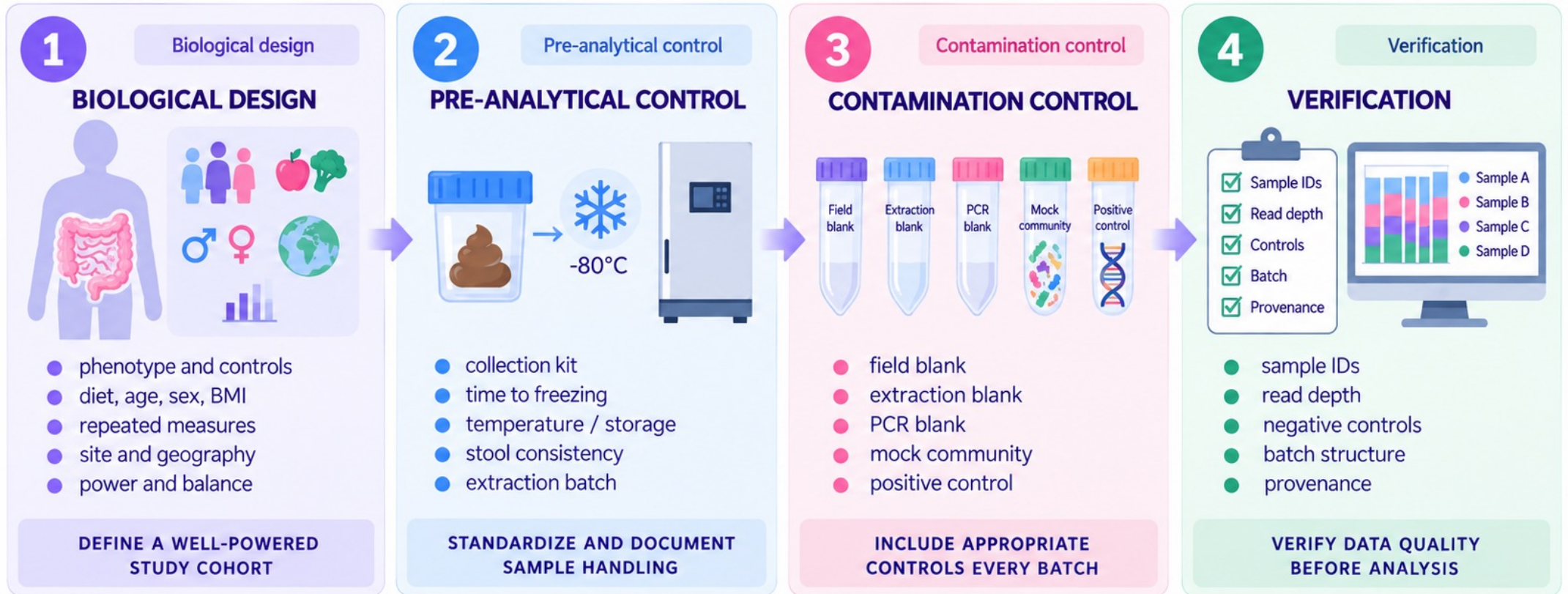


Measure the biological layer you **intend to interpret**.

Microbiome Study Design Begins Before Sequencing

Sampling, metadata, contamination controls, and batch structure shape the observed community.

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Low-biomass samples require stronger contamination evidence because reagent DNA may exceed true microbial biomass.



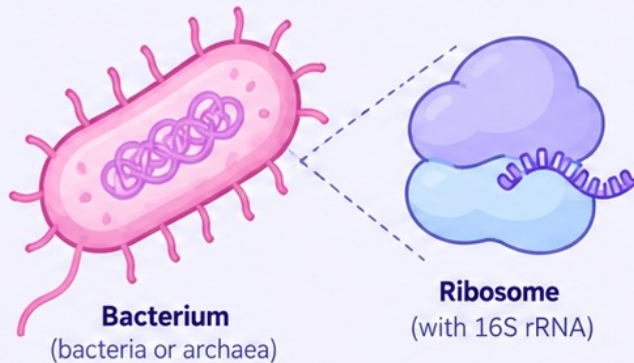
A strong model **cannot rescue** a confounded or contaminated experiment.

Why 16S Works as a Microbial Marker

Conserved regions support primer binding; variable regions provide taxonomic information.

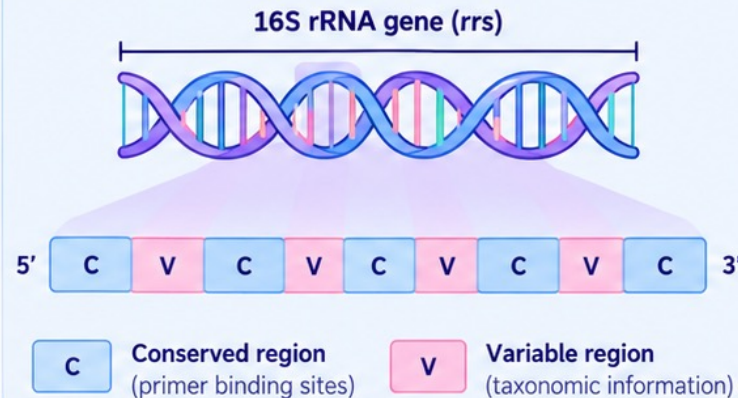
From cells to sequences

Bacteria and archaea contain ribosomes with 16S rRNA, encoded by the 16S rRNA gene (*rrs*) in the genome.



16S rRNA gene structure

The 16S rRNA gene contains conserved regions that are similar across taxa and variable regions that accumulate taxonomic differences.



What 16S can measure



- bacterial / archaeal membership
- community comparison
- ecological screening

What 16S cannot measure



- whole genomes
- most viruses and fungi
- activity or metabolites
- causality



Interpretation discipline

16S measures marker sequences, not complete organisms or function.



16S identifies marker sequences, **not complete genomes.**

The Selected 16S Region Changes What You Can See

Region choice affects resolution, bias, cross-study comparability, and cost.

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V4



- ~250 bp
- excellent paired-end overlap
- strong cross-study comparability
- lower species resolution

V3–V4



- ~460 bp
- more marker context
- common gut-survey design
- region-specific bias

Full-length V1–V9



- ~1.5 kb
- long-read marker context
- better discrimination for some taxa
- primer and copy-number bias remain

Shotgun metagenomics



- genome-wide DNA
- no 16S-region choice
- species/strain + genes/pathways
- more cost, host DNA, and compute

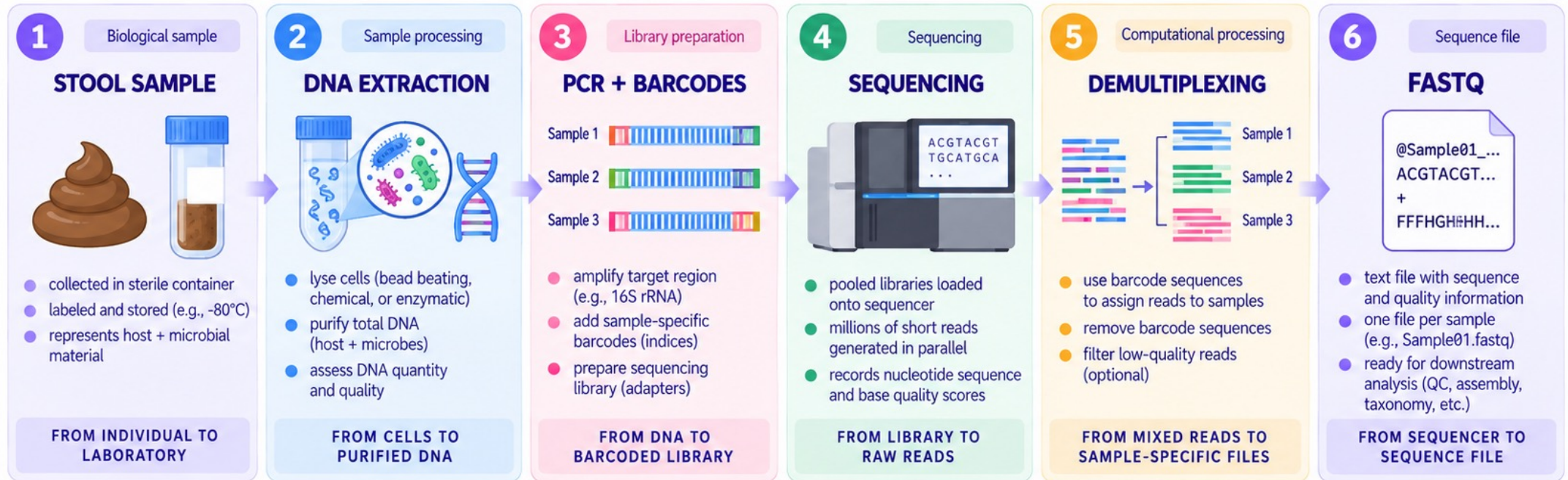


More sequence context improves resolution, but it **does not remove** primer or copy-number bias.

From a Stool Sample to FASTQ Files

Biological material becomes a sequence file through a traceable laboratory workflow.

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What FASTQ contains

Each read in a FASTQ file consists of four lines:

- 1 read identifier** Unique ID for the sequencing read (e.g., instrument, lane, coordinates).
- 2 nucleotide sequence** The DNA sequence (A, C, G, T, and sometimes N).
- 3 base quality** ASCII-encoded quality scores for each base call (e.g., Phred scores).

Example FASTQ record

```
1 @Sample01_read0001  
2 ACGTACGTACGT...  
3 +  
4 FFFHGH#HHHHH...
```

- Read identifier**
e.g., instrument:lane:tile:x:y
- Nucleotide sequence**
A, C, G, T (and sometimes N)
- Separator line**
Always a "+"
- Base quality scores**
ASCII-encoded (e.g., Phred+33)



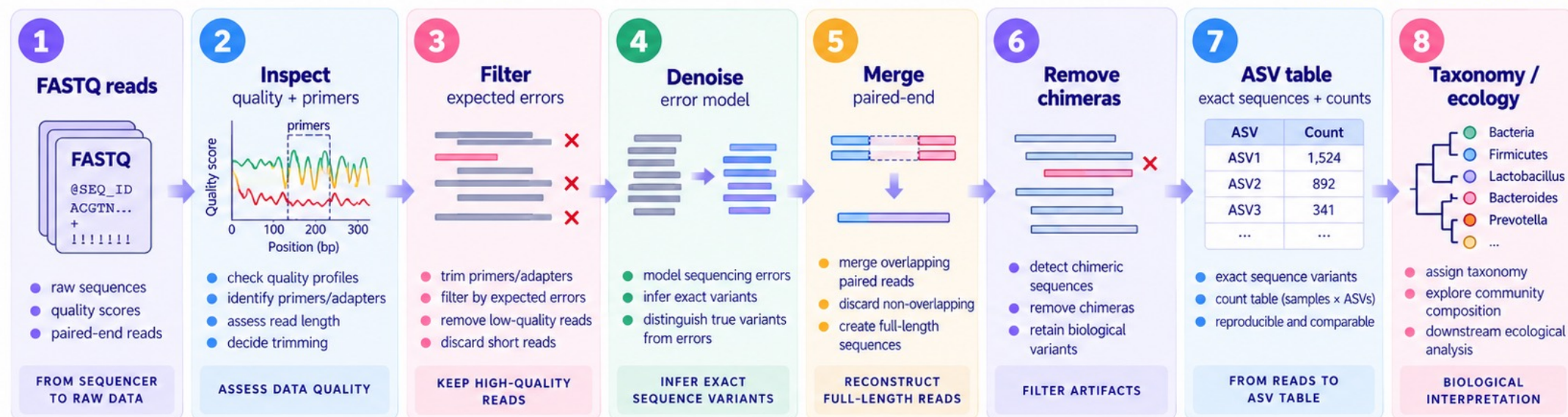
FASTQ is the first computational representation of the biological sample.

From FASTQ Reads to ASVs

Denoising converts raw reads into exact sequence variants with a quality trail.

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Mandatory read-retention trail

Track how many reads are retained at each step to ensure transparency and data quality.



- ✓ Monitor read loss at each step
- ✓ Identify potential problems (e.g., low merge rate)
- ✓ Ensure reproducibility

Outputs to save

Keep key files for reproducibility, downstream analysis, and re-use.

- quality profiles
- ASV sequences + count table
- taxonomy + confidence
- denoising / merge statistics
- parameters + version information

- ✓ Enable downstream analyses
- ✓ Facilitate methods comparison
- ✓ Support re-analysis and re-use
- ✓ Document analysis settings
- ✓ Share with the community



ASVs are inferred sequences—not automatically species or strains.

Taxonomy Is an Inference, Not Ground Truth

A sequence is mapped into a versioned reference system using a chosen classifier.

Query sequence

ASV / amplicon sequence



Classifier

- region-specific training
- confidence threshold
- algorithm + parameters



Reference system



- SILVA SSU 144
- Greengenes2 / genome-linked phylogeny
- other curated databases

Report with every taxonomic result



Database
name + release



Region
e.g. V4 / V3-V4



Classifier
method + confidence



Unclassified
retain and report



Resolution
avoid unsupported
species claims



A taxonomic label is **reproducible only when** the sequence, database, version, and classifier are reproducible.

Anatomy of a Microbiome Dataset

A microbiome dataset is more than one table: features, samples, metadata, taxonomy, and phylogeny must stay linked.

1

Taxonomy table

Maps each feature (OTU/ASV) to a taxonomic lineage.

Feature ID	Genus	Family	Phylum
OTU1	Bacteroides	Bacteroidaceae	Bacteroidota
OTU2	Faecalibacterium	Ruminococcaceae	Firmicutes
OTU3	Bifidobacterium	Bifidobacteriaceae	Actinomycetota
...	...	...	...

Kingdom

Phylum

Class

Order

Family

Genus

Species

Feature abundance matrix

Counts or relative abundances of features in each sample (e.g., OTUs or ASVs).



Rows = features; columns = samples;
IDs must match across files.

2

Sample metadata

Describes each sample and its experimental context (e.g., phenotype, demographics, study design).

Sample ID	Group	Age	Sex	BMI	Other
S1	Control	34	F	22.1	Site A
S2	Treatment	28	M	27.4	Site A
S3	Disease	52	F	31.2	Site B
...	...	...	...	...	...

3

Sequence data

Nucleotide sequences for each feature (OTU/ASV), used for taxonomic assignment and phylogenetic analysis.

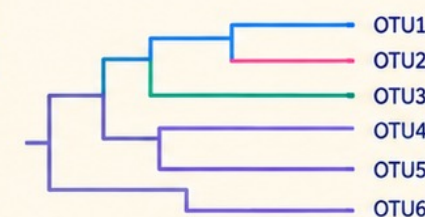
Feature ID	Representative sequence (5' → 3')
OTU1	ACGTTGACCTGATCGGA...
OTU2	TTAACGGTAGCCATGTT...
OTU3	GGTTACCTGAACTGCC...
...	...



4

Phylogenetic tree

Shows evolutionary relationships among features, often built from the sequence data.



- Built from aligned sequences
- Used in diversity and compositional methods
- Branch lengths reflect evolutionary distance



Count table

Feature × sample matrix
(OTUs/ASVs × samples)



Taxonomy

Links each feature to a taxonomic lineage



Metadata

Describes each sample and its context



Phylogeny

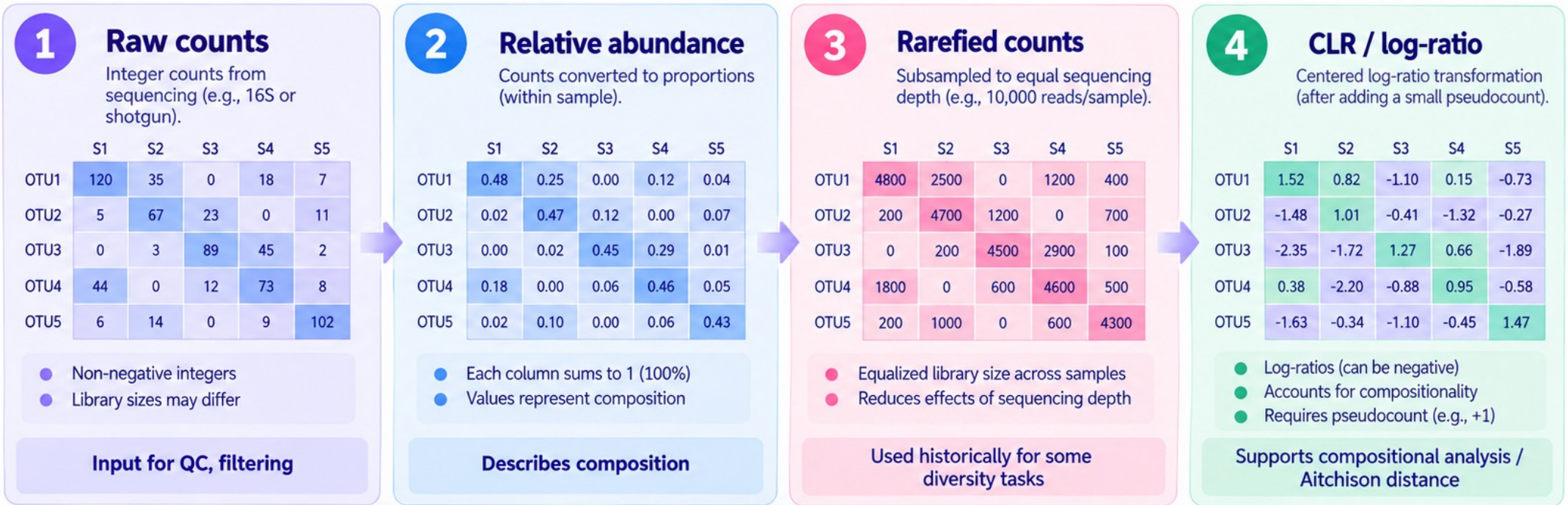
Evolutionary relationships among features



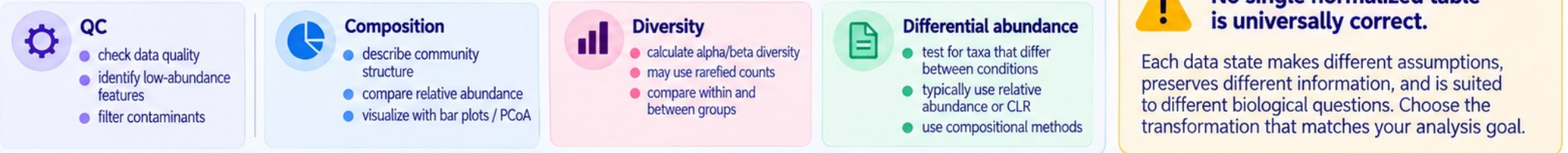
A microbiome analysis begins with a **linked data object**, not a single spreadsheet.

The Same Dataset Exists in Different Data States

Microbiome count data can be transformed into different analysis-ready forms, and each state supports a different question.



Choose the data state to match the question



The same samples can yield **different valid data states** for different analyses.

Relative Abundance Is Not Absolute Abundance

Because sequencing data are closed to a total, one taxon can appear to change even when others stay constant.

Absolute cells: Sample A

All three taxa have the same number of cells in the sample.

Total cells
300



Absolute cells: Sample B

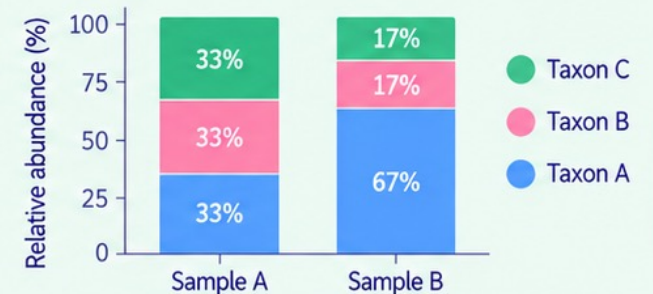
Taxon A has increased in absolute abundance, while Taxon B and C remain the same.

Total cells
600



Observed proportions after closure

Sequencing measures relative abundance, so the increase in Taxon A makes the proportions of B and C appear smaller.



INTERPRETATION

Taxon B and Taxon C did not decrease in absolute abundance; they only decreased in relative proportion.

- ✓ Taxon A truly increased in absolute abundance (100 → 400).
- ✓ Taxon B and Taxon C stayed constant (100 → 100).
- ✓ Because sequencing data are closed to a total, the proportions of B and C must decrease when A increases.

Methods to estimate absolute abundance

Combine sequencing with independent quantification methods.



qPCR

Targeted quantification of specific taxa



Spike-in

Known quantities of external standards



Flow cytometry

Total cell counts and cell sorting



Biomass

Total microbial biomass (e.g., dry weight, ATP)

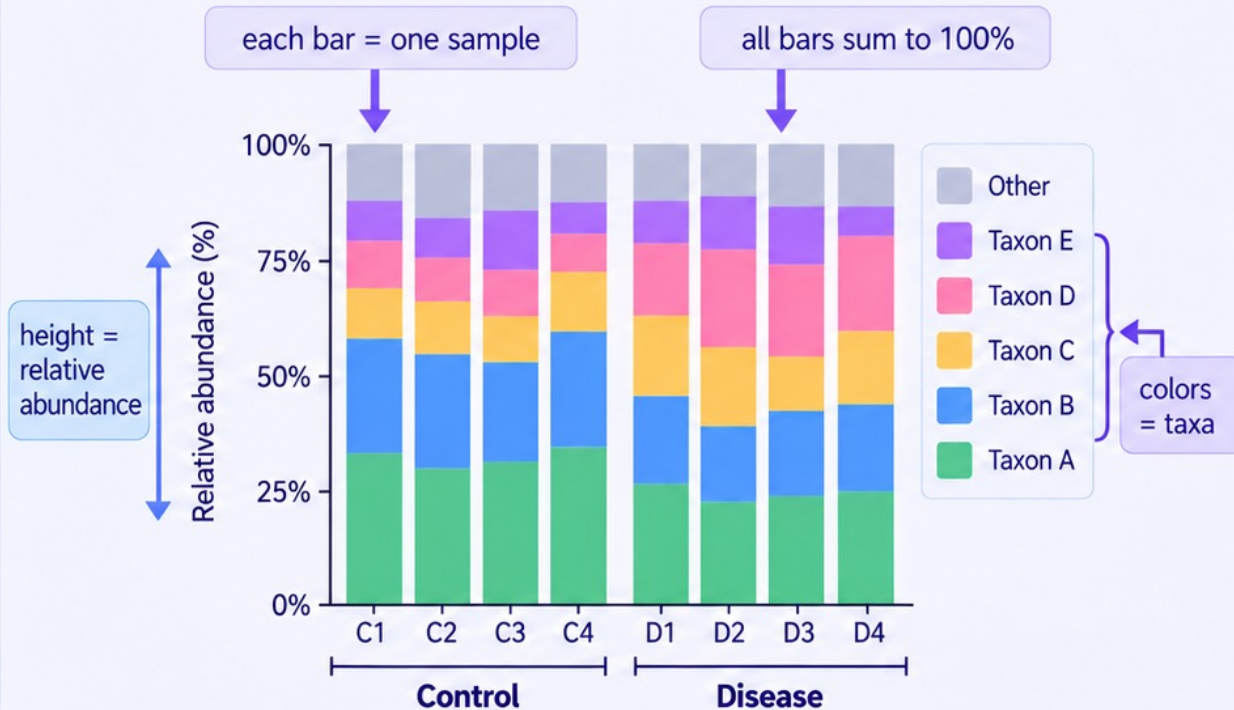


Relative sequencing alone **cannot prove** absolute expansion or depletion.

How to Read a Taxonomic Bar Plot

Stacked bar plots summarize composition, but they do not test significance by themselves.

Example: gut microbiome composition



What you **can** say

- community composition differs visually
- dominant taxa can be compared descriptively
- sample-to-sample heterogeneity is visible



What you **cannot** say

- Taxon X is significantly different
- Taxon Y decreased in absolute abundance
- the mechanism is known



Filtering and the definition of 'Other' affect the plot.



A bar plot is descriptive composition, **not statistical inference**.

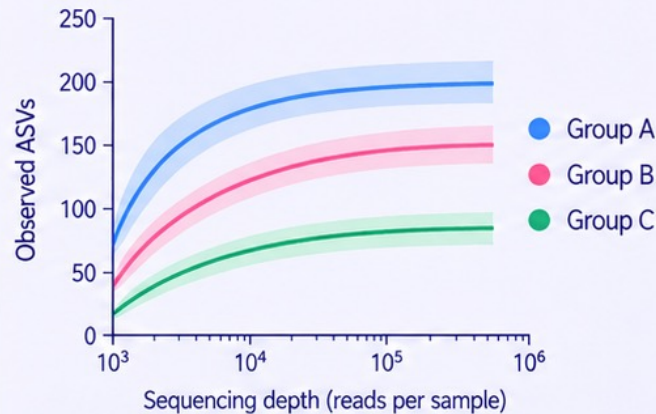
Diversity Metrics Summarize Different Ecological Properties

Alpha diversity describes within-sample diversity; beta diversity compares samples to one another.

A

Sampling depth and richness

Rarefaction curves show how observed ASVs increase with sequencing depth and can indicate whether sampling captured most of the diversity.

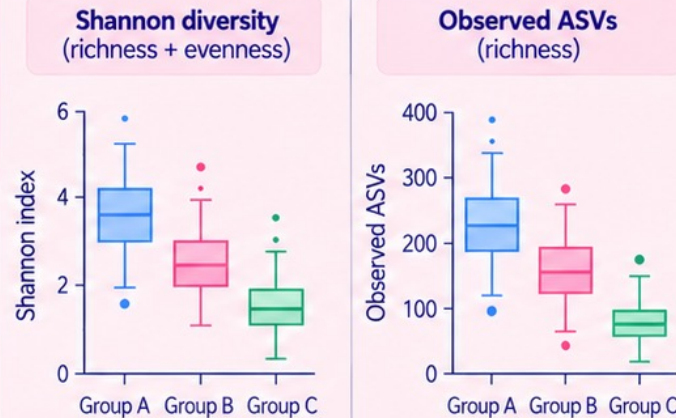


Compare sampling depth and richness across groups to ensure fair comparisons.

B

Alpha diversity

Alpha diversity metrics summarize within-sample diversity, combining richness and evenness.

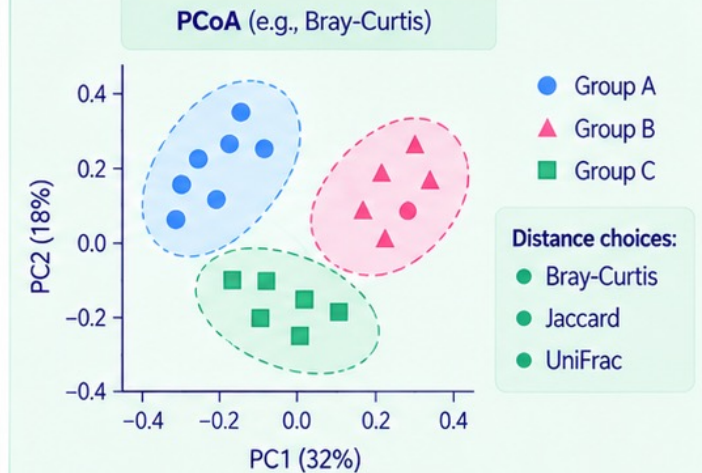


Alpha diversity tests whether samples differ in within-sample diversity.

C

Beta diversity + inference

Beta diversity compares community composition between samples using a distance matrix.



Test both centroid separation (PERMANOVA) and dispersion.



Key lesson

1

Check sampling depth and richness first.

2

Use alpha diversity to compare within-sample diversity.

3

Use beta diversity and statistical tests to compare samples.



A PCoA plot is a projection—not a significance test, diagnostic label, or proof of discrete community types.

What is microbial richness and evenness?

Two complementary aspects of alpha diversity

1

Richness

How many different microbial taxa are present in a sample?

Richness is the **number of distinct taxa** (e.g., species/OTUs/ASVs) observed in a sample.

Low richness



2 taxa

Community with fewer different taxa.

High richness



8 taxa

Community with more different taxa.



Common richness metrics

- Observed species (OTUs/ASVs): count of detected taxa
- Chao1: estimated total richness (including unseen taxa)



Key point

Richness focuses on the number of taxa, regardless of their abundance.

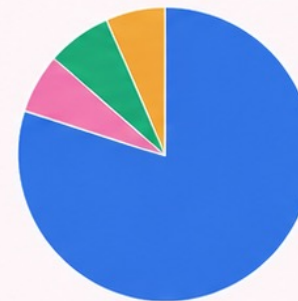
2

Evenness

How evenly are the taxa distributed in abundance?

Evenness describes **how similar the abundances** of different taxa are in a sample.

Low evenness



- Taxon 1 80%
- Taxon 2 5%
- Taxon 3 5%
- Taxon 4 5%
- Taxon 5 5%

One taxon dominates

Low evenness: most reads belong to one taxon.

High evenness



- Taxon 1 20%
- Taxon 2 20%
- Taxon 3 20%
- Taxon 4 20%
- Taxon 5 20%

Taxa are similarly abundant

High evenness: taxa have similar abundances.



Common evenness metrics

- Pielou's evenness (J): derived from Shannon index
- Shannon index (H'): considers both richness and evenness
- Simpson index (D): more sensitive to dominant taxa



Key point

Evenness focuses on the distribution of abundances, regardless of the number of taxa.



Take-home message: Richness tells you how many different microbes are present, and evenness tells you how evenly they are distributed. Both are important and complementary measures of alpha diversity, and should be interpreted together.

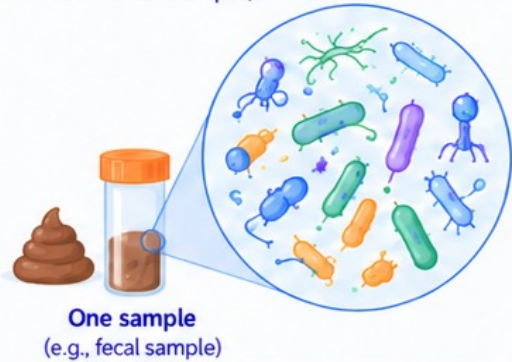
Alpha diversity: within-sample microbial diversity

How diverse is the microbiome in a single sample?

Alpha diversity measures the richness and evenness of microbial communities within a single sample.

A What is alpha diversity?

Describes the diversity of microorganisms within **one sample** (e.g., one individual, one environmental sample).



Alpha diversity considers both the **number of species (richness)** and their **relative abundances (evenness)**.

B Common alpha diversity metrics



Observed richness

Number of observed species (OTUs/ASVs)

S = number of species



Chao1 (richness estimator)

Estimates total species richness, including unseen species

$$\text{Chao1} = S_{\text{obs}} + \frac{F_1^2}{2F_2}$$

F_1 : singletons
 F_2 : doubletons



Shannon index

Considers both richness and evenness

$$H' = - \sum_{i=1}^S p_i \ln(p_i)$$



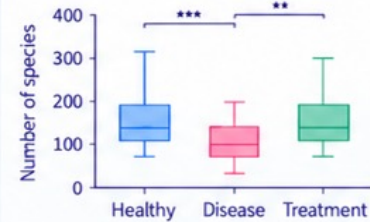
Simpson index

Measures the probability that two randomly selected individuals are different

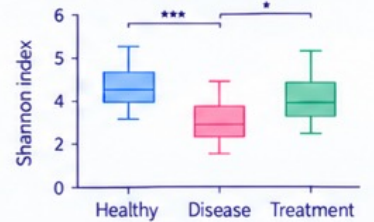
$$D = 1 - \sum_{i=1}^S p_i^2$$

C Example plots and interpretation

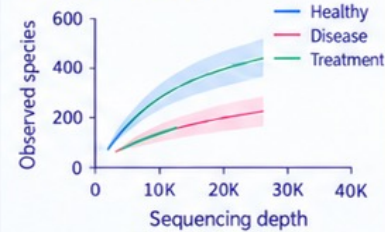
Observed richness



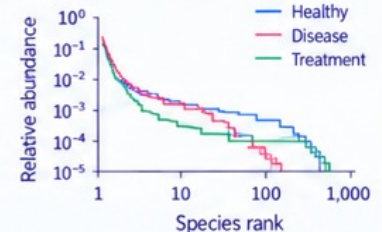
Shannon index



Rarefaction curve



Rank-abundance curve



D How to interpret alpha diversity?

Higher alpha diversity may indicate:

- ✓ A more diverse and stable ecosystem
- ✓ Better resilience to perturbation
- ✓ Often observed in healthy individuals (in many, but not all, contexts)

Lower alpha diversity may indicate:

- ✓ Reduced microbial diversity
- ✓ Potential dysbiosis
- ✓ Often observed in disease states (e.g., IBD, obesity, antibiotic use)

E Factors affecting alpha diversity



Diet
(fiber, fat, etc.)



Antibiotics
and drugs



Host factors
(age, genetics,
health status)



Environment
(geography, lifestyle)



Sampling and
technical factors
(depth, method)



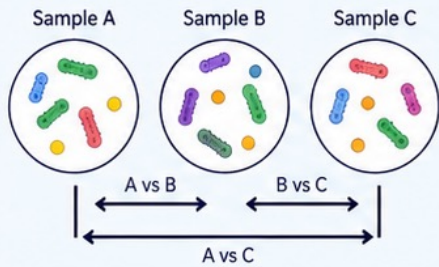
Take-home message: Alpha diversity quantifies the richness and evenness of microbial communities within a sample, providing key insights into ecosystem health and stability, but needs to be interpreted together with beta diversity and taxonomic/functional information.

Beta diversity: how different are microbial communities between samples?

Measure between-sample community dissimilarity and interpret ordination with statistical caution

1 What is beta diversity?

Beta diversity refers to differences in community composition between samples.

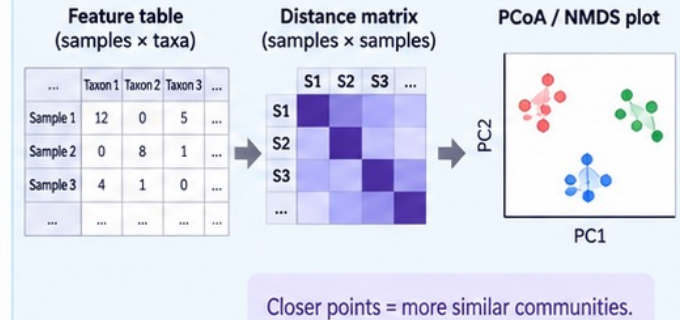


- between-sample diversity
- uses a distance / dissimilarity matrix
- asks whether communities are similar or different

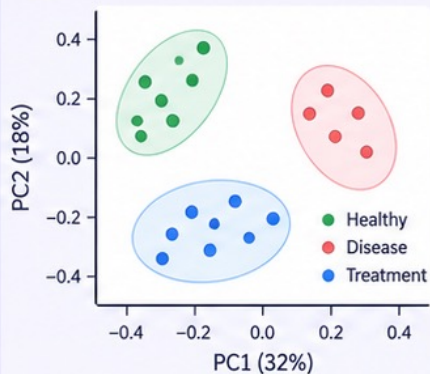
2 Common distance metrics

Jaccard	Presence / absence only Captures differences in community membership (which taxa are present).
Bray–Curtis	Abundance-based Sensitive to differences in relative abundances of shared taxa.
Unweighted UniFrac	Phylogeny + presence/absence Captures phylogenetic turnover in community membership.
Weighted UniFrac	Phylogeny + abundance Incorporates phylogenetic relationships and relative abundances.
Aitchison distance	Compositional / log-ratio framework Designed for compositional data; captures differences in relative abundances (log-ratios).

3 From distance matrix to ordination



4 How to read a PCoA plot



- Each dot = one sample
- Distance between dots reflects community dissimilarity
- Axis percentages indicate explained variation
- Visual separation is descriptive, not a formal test

5 Statistics and interpretation

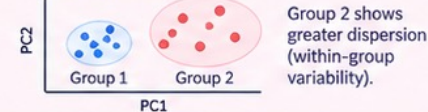
A PERMANOVA

Tests whether group centroids differ (i.e. community composition differs between groups).

$$R^2 = 0.12, P = 0.003$$

B Dispersion check (betadisper / PERMDISP)

Tests whether within-group variability (dispersion) differs between groups.



Interpretation guide

Significant PERMANOVA + similar dispersion	→ Group composition differs between samples.
Significant PERMANOVA + unequal dispersion	→ Interpret with caution (differences may be driven by dispersion).

6 Key caveats

- ⚠ Results depend on the chosen metric.
- ⚠ Sequencing depth, normalization, and rare taxa can affect distances.
- ⚠ Beta diversity does not identify which taxa drive the difference.
- ⚠ Separation does not imply causality.
- ⚠ Always pair with differential abundance and metadata analysis.



Take-home message: Beta diversity compares microbial community composition between samples; choose the right distance metric, visualize with ordination, and confirm differences with PERMANOVA and dispersion checks.

How to choose a beta diversity metric

Start from the biological question, then match the metric to the type of community difference you want to detect.

1 Start with the biological question



Who is present or absent?



How do abundances differ?



Do phylogenetic lineages differ?



Do relative abundance ratios change?



Do I care about rare taxa or dominant taxa?

The question should determine the metric,
not the other way around.

2 Metric selection map

Metric	Captures best	Best when	Main limitation
Jaccard	Presence / absence of taxa	Gain or loss of taxa between samples	Ignores abundance information
Bray–Curtis	Abundance differences	Comparing relative abundances between samples	No phylogenetic information
Unweighted UniFrac	Phylogeny + presence/absence	Lineage turnover and rare taxa	Sensitive to low-abundance detection
Weighted UniFrac	Phylogeny + abundance	Dominant lineage shifts	Influenced by abundant taxa
Aitchison distance	Compositional log-ratio changes	Compositional analysis	Requires thoughtful zero handling

3 Practical decision flow

Only care whether taxa are present or absent?



Jaccard

Care about abundance differences?



Bray–Curtis

Does phylogeny matter, and presence/absence is enough?



Unweighted UniFrac

Does phylogeny matter, and abundance matters?



Weighted UniFrac

Want a compositional / log-ratio framework?



Aitchison distance

4 Common study scenarios

Antibiotic treatment causes loss of taxa



Jaccard / Unweighted UniFrac

Disease alters dominant community structure



Bray–Curtis / Weighted UniFrac

Ecological turnover across environments



Unweighted UniFrac / Weighted UniFrac

Shotgun species table with compositional focus



Aitchison / Bray–Curtis

5 Selection standards

- ✓ Match the metric to the biological question.
- ✓ Consider whether abundance matters.
- ✓ Consider whether phylogeny matters.
- ✓ Check sequencing depth and sparsity.
- ✓ Use the same preprocessing across samples.
- ✓ Test robustness with more than one metric when possible.
- ✓ Always document metric choice and rationale.

6 Interpretation discipline



Different metrics can produce different ordination patterns.



A metric does not remove confounding or batch effects.



PCoA separation still requires PERMANOVA and dispersion checks.



No single metric is universally best.



Explain why the chosen metric is biologically appropriate.



Take-home message: Choose beta diversity metrics by the biological question—presence/absence, abundance, phylogeny, or compositional structure—and confirm that conclusions are robust and well justified.

How to Read a PCoA Plot

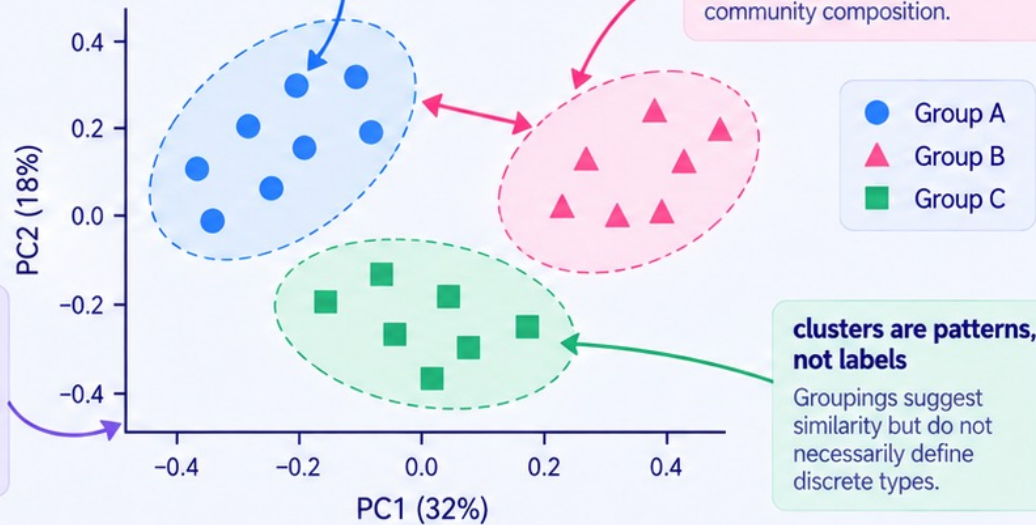
A PCoA plot displays distances among samples in a low-dimensional projection; each point is one sample.

Example PCoA plot

Each point represents one sample, and distances reflect compositional dissimilarity (e.g., Bray-Curtis, UniFrac).

one dot = one sample
Each point represents an individual sample.

distance reflects compositional dissimilarity
Points that are farther apart have more different community composition.



axes summarize major variation

PC1, PC2, etc. capture the largest sources of variation in the data (percentages show explained variation).

clusters are patterns, not labels

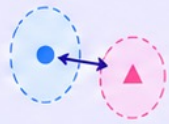
Groupings suggest similarity but do not necessarily define discrete types.

Two common comparisons

PCoA plots can suggest different patterns of centroids (group means) and dispersion (within-group spread).

Centroid difference

Groups may have different average community composition (centroids), indicating separation.



Dispersion difference

Groups may differ in within-group spread (dispersion), even if centroids are similar.



What you can read

- **sample similarity / separation**
Points that are closer are more similar in community composition.
- **possible group structure**
Clusters suggest shared features among samples.
- **outliers**
Samples that fall far from others may be unique.
- **within-group spread**
The spread of points indicates the heterogeneity within a group.

What you cannot claim from the plot alone

- **statistical significance**
A PCoA plot does not provide p-values.
- **mechanism**
It does not reveal which features drive the differences.
- **causality**
It cannot establish cause-and-effect relationships.
- **discrete biological types**
Clusters are patterns, not proof of distinct taxa or states.



A PCoA plot is a projection, **not a statistical conclusion.**

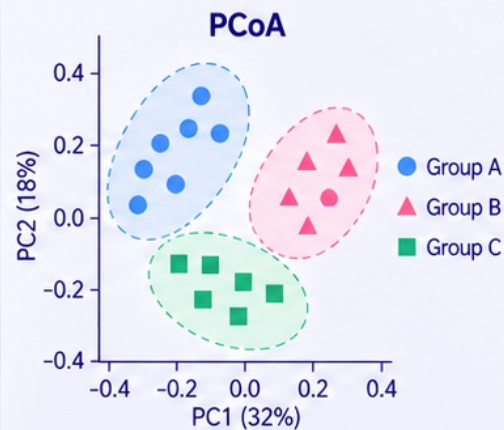
PCoA Is a Picture; PERMANOVA Is the Test

Visual separation should be checked with a distance-based statistical test and with a dispersion assessment.

1

Ordination view

PCoA visualizes sample relationships in a low-dimensional space based on a distance matrix.

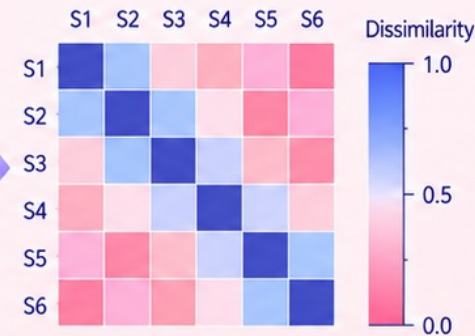


visual pattern

2

Distance matrix

Samples are compared using a distance matrix (e.g., Bray-Curtis, UniFrac), which quantifies pairwise dissimilarities.



pairwise dissimilarities
among samples

3

Statistical inference

Use distance-based tests to evaluate whether groups differ and to check for differences in dispersion.

PERMANOVA
 $R^2 = 0.08$, $P = 0.002$

Dispersion test (e.g., betadisper)
 $P = 0.62$

Interpretation
Group explains ~8% of
between-sample variation
after accounting for dispersion.



Interpretation discipline

Follow these steps to make robust conclusions from ordination analyses.

test centroid separation
(e.g., PERMANOVA)

check dispersion
(e.g., betadisper)

report effect size + P value
(e.g., R^2 and P)

translate statistics into plain biology
(biological meaning and context)



Key lesson

1

Use the figure to see
the pattern.

2

Use a distance-based test
and check dispersion.

3

Interpret effect size and P value
in biological context.



Use the figure to **visualize the pattern**, and the test to **evaluate the evidence**.

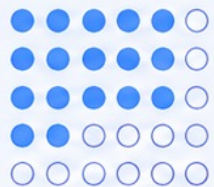
Differential Abundance: What Exactly Changed?

Before choosing a method, define the quantity that differs across conditions.

1 Prevalence change

A taxon is detected in a different fraction of samples across conditions (present vs. absent).

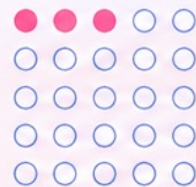
Control (higher prevalence)



8/10 samples (80%)

● Detected ○ Not detected

Disease (lower prevalence)



3/10 samples (30%)

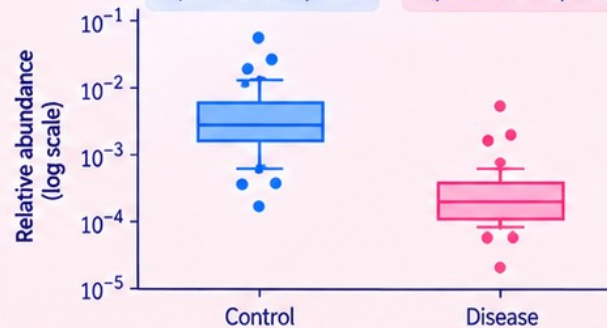
● Detected ○ Not detected

Detected in fewer samples.

2 Abundance change

A taxon is present in both conditions, but its abundance differs among positive samples.

Control
(positive samples)



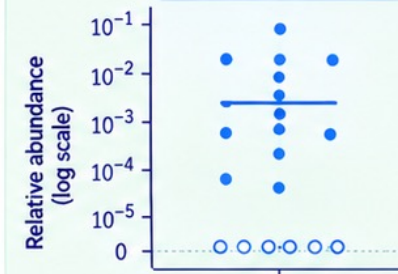
Amount among positive samples differs.

Disease
(positive samples)

3 Both changed

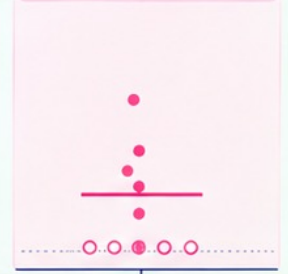
A taxon is detected in fewer samples and, among those where it's detected, it has lower abundance.

Control



8/10 detected (80%)

Disease



3/10 detected (30%)

Detected in fewer samples and lower abundance among positive samples.

Typical analysis goals



Screen candidate taxa

Identify features that differ across conditions.



Estimate effect size

Quantify the magnitude and direction of change (e.g., log fold change).



Adjust for covariates

Account for confounders such as age, batch, or sequencing depth.



Control false discoveries

Use multiple testing correction to limit false positives.

Example methods

ANCOM-BC2
MaAsLin 3
ALDEx2
LinDA



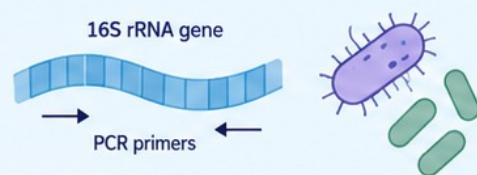
Ask **what quantity changed** before asking which package found it.

Taxonomic resolution differs across 16S, 18S, ITS, and shotgun metagenomics

Marker choice determines which organisms you see and how precisely you can classify them.

16S rRNA amplicon

Bacterial and archaeal community profiling



Target groups: Bacteria + Archaea

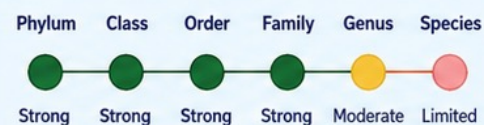


Typical best use: Community surveys, microbiome profiling



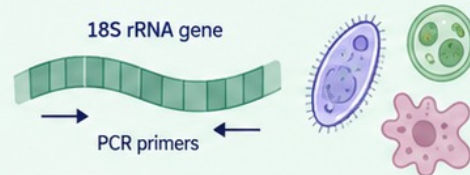
Notes: Primer and region choice matter.

Typical taxonomic resolution:



18S rRNA amplicon

Microbial eukaryote community profiling



Target groups: Microbial eukaryotes (protists, algae, etc.)

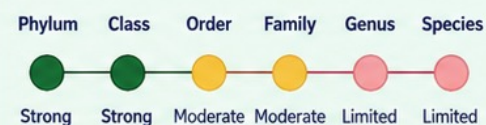


Typical best use: Broad eukaryotic community analysis



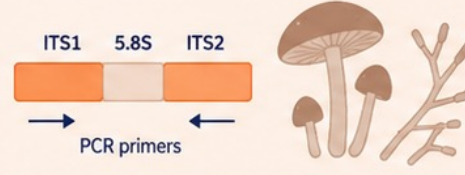
Notes: Lower species-level resolution compared to other markers.

Typical taxonomic resolution:



ITS amplicon

Fungal community profiling



Target groups: Fungi

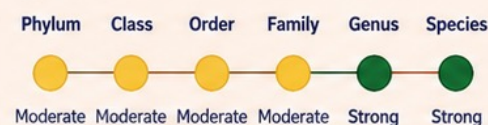


Typical best use: Fungal community profiling (e.g., environmental, host-associated)



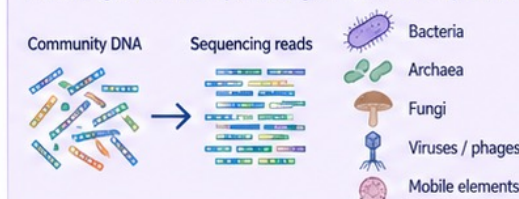
Notes: Preferred marker for fungal profiling.

Typical taxonomic resolution:



Shotgun metagenomic sequencing

Whole-genome sequencing of community DNA



Target groups: Bacteria, Archaea, Fungi, Viruses/Phages, and mobile elements

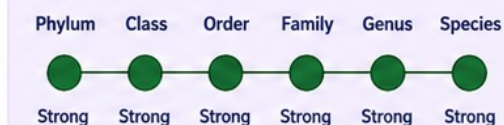


Typical best use: Comprehensive taxonomic and functional profiling



Notes: Provides genes/pathways; can reach strain-level when depth and references are sufficient.

Typical taxonomic resolution:



What to conclude

- 1 16S is excellent for bacterial/archaeal community surveys.
- 2 18S broadens analysis to microbial eukaryotes.
- 3 ITS is the preferred marker for fungal profiling.
- 4 Shotgun generally offers finer taxonomic resolution and broader functional insight.



Important caveats

- 1 Resolution depends on database quality and read length.
- 2 Marker sequencing usually cannot support confident species calls for all taxa.
- 4 Shotgun is more informative but requires greater sequencing depth and cost.



Amplicon markers are powerful for targeted community profiling, whereas shotgun metagenomics usually provides broader coverage and finer taxonomic resolution.

16S or Shotgun? Choose the Assay by the Question

Use the least complex assay that can answer the biological question with adequate controls.

16S / marker gene



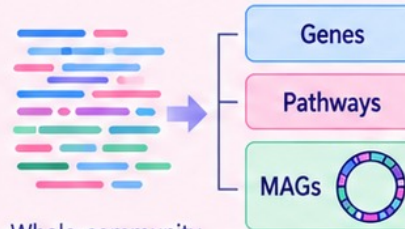
16S rRNA gene
(e.g., V3–V4)

- **primary signal:** selected marker region
- **typical resolution:** genus; species when supported
- **functional inference:** indirect and uncertain
- **major biases:** PCR, primers, copy number
- **strength:** cost-effective community surveys
- **best use:** large cohorts, ecology, screening

TARGETED VIEW OF COMMUNITY COMPOSITION

FAST. COST-EFFECTIVE. BROAD SURVEYS.

Shotgun metagenomics



Whole-community
DNA sequencing

- **primary signal:** all recoverable DNA
- **typical resolution:** species/strain when supported
- **functional evidence:** genes, pathways, MAGs
- **major biases:** host DNA, extraction, database
- **strength:** taxonomy + function + MAGs
- **best use:** mechanism, resistome, strain tracking

COMPREHENSIVE VIEW OF COMMUNITY POTENTIAL

TAXONOMY + FUNCTION + STRAINS. HIGHER COMPLEXITY.

Escalate only when the question requires it:



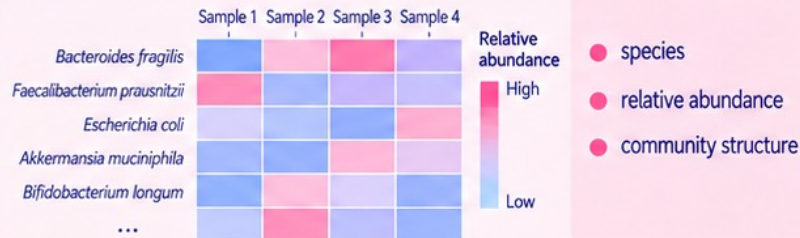
Neither assay directly measures viability, transcription, protein activity, metabolite flux, or absolute abundance.

Shotgun Metagenomics Creates Multiple Data Matrices

The same sequencing reads can be represented as different analysis-ready objects, each answering a different question.

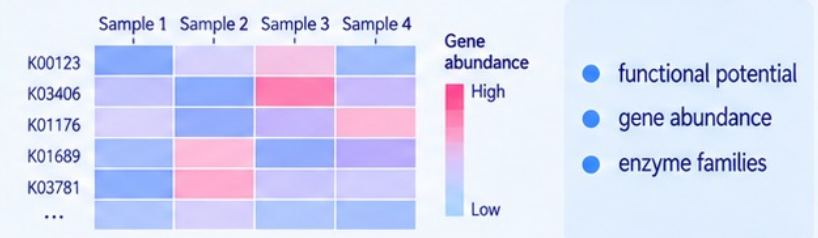
1 Taxonomic profile

Species-by-sample abundance matrix



2 Gene-family table

Genes-by-sample abundance matrix



Shotgun FASTQ reads

```
@SEQ_ID
ACGTTGCAATGCC...
+
IIIIIIIIII....
```

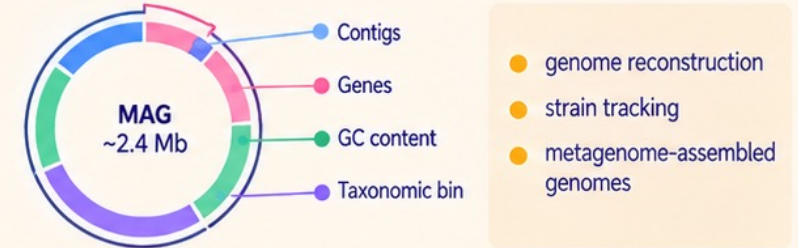
3 Pathway table

Pathway-by-sample abundance matrix



4 Strain / MAG data

Genome-level objects from the same reads



Key lesson

1

Same reads

One sequencing experiment (shotgun FASTQ files).

2

Different matrices

The same reads can be processed into multiple analysis-ready objects.

3

Different biological claims

Each matrix answers a different biological question about the community.



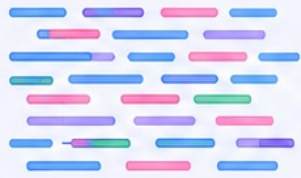
Shotgun metagenomics is **not one dataset**; it is a family of related data representations.

Read-Level Profiling vs Genome Reconstruction

Shotgun metagenomics supports two related but different analysis branches.

Shotgun FASTQ reads

Millions of short reads from a microbial community.



```
@SEQ_ID
ACGTTGCAATGCC...
+
IIIIIIIIIIIIIIIIIIII...
```

Host removal

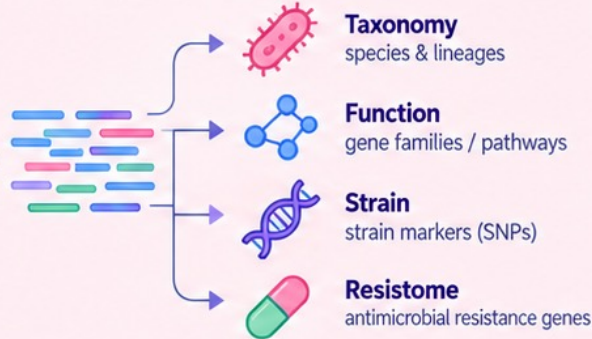
Map to host genome and remove human/host reads.



Keep non-host (microbial) reads.

Read-level profiling

Map reads directly to reference databases to quantify what is present.

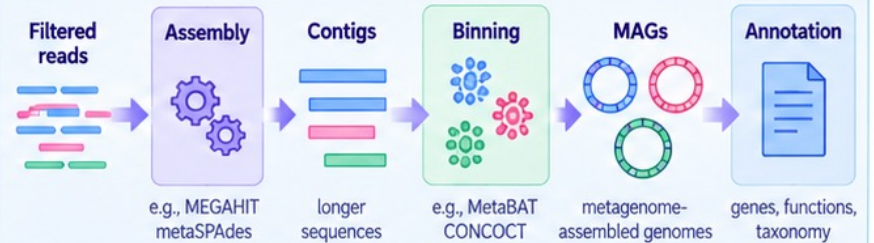


Asks: what is detected?

- species abundance
- gene families / pathways
- strain markers
- no assembly required

Assembly-based reconstruction

Assemble reads into longer sequences, bin into genomes, and annotate resulting MAGs.



Asks: which genomes can be reconstructed?

- genome-level objects
- completeness / contamination checks
- taxonomy + genes
- ecological validation

Profiling

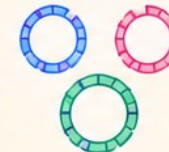
Read-level analysis is fast, sensitive, and quantitative, but does not reconstruct genomes.



- ✓ Relative abundance
- ✓ Functions / pathways
- ✓ Strain-level signals
- ✗ No genome reconstruction

Assembly

Assembly reconstructs longer sequences and genome-level objects, enabling deeper analysis, but is more computationally intensive.



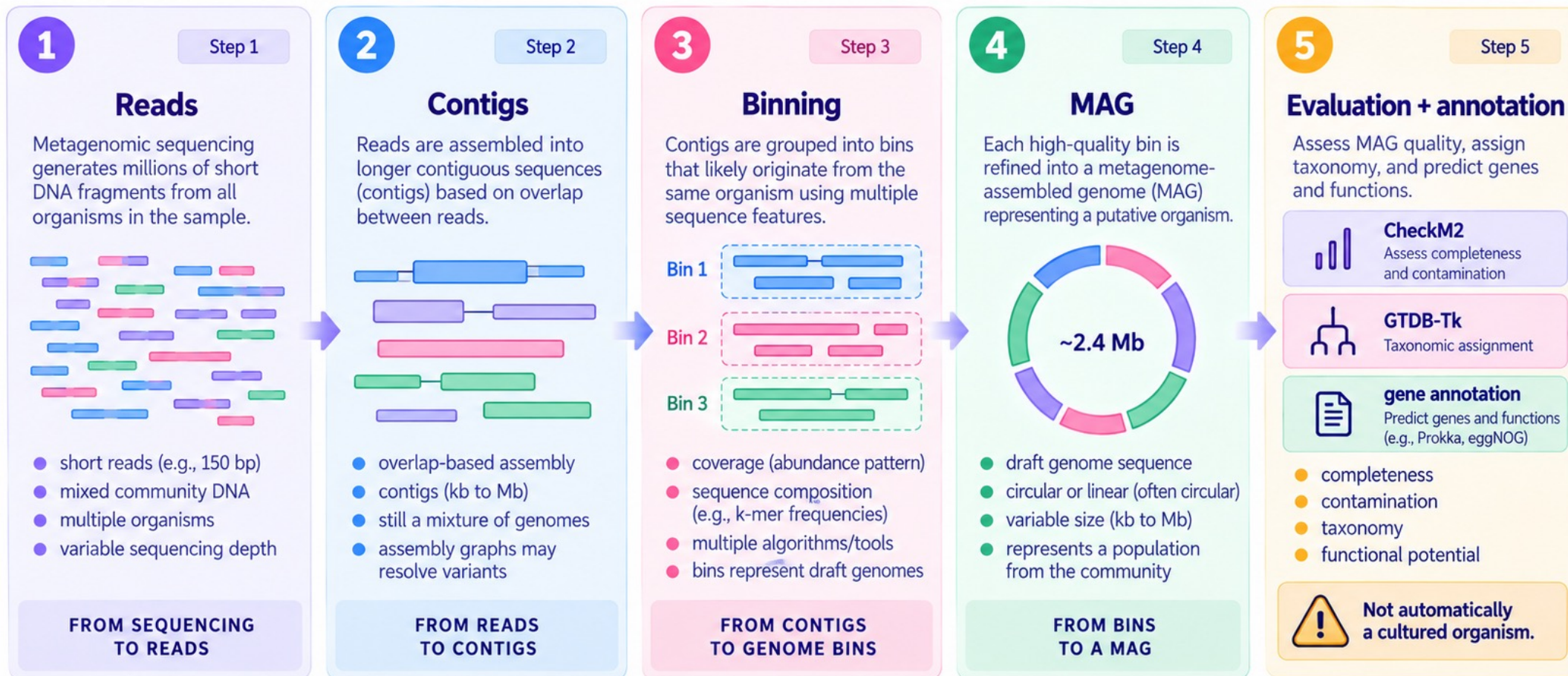
- ✓ Genome-level MAGs
- ✓ Completeness / contamination
- ✓ Taxonomy + genes
- ✗ Higher computational cost



Read profiling asks **what is detected**; assembly asks **which genomes can be reconstructed**.

How a MAG Is Reconstructed

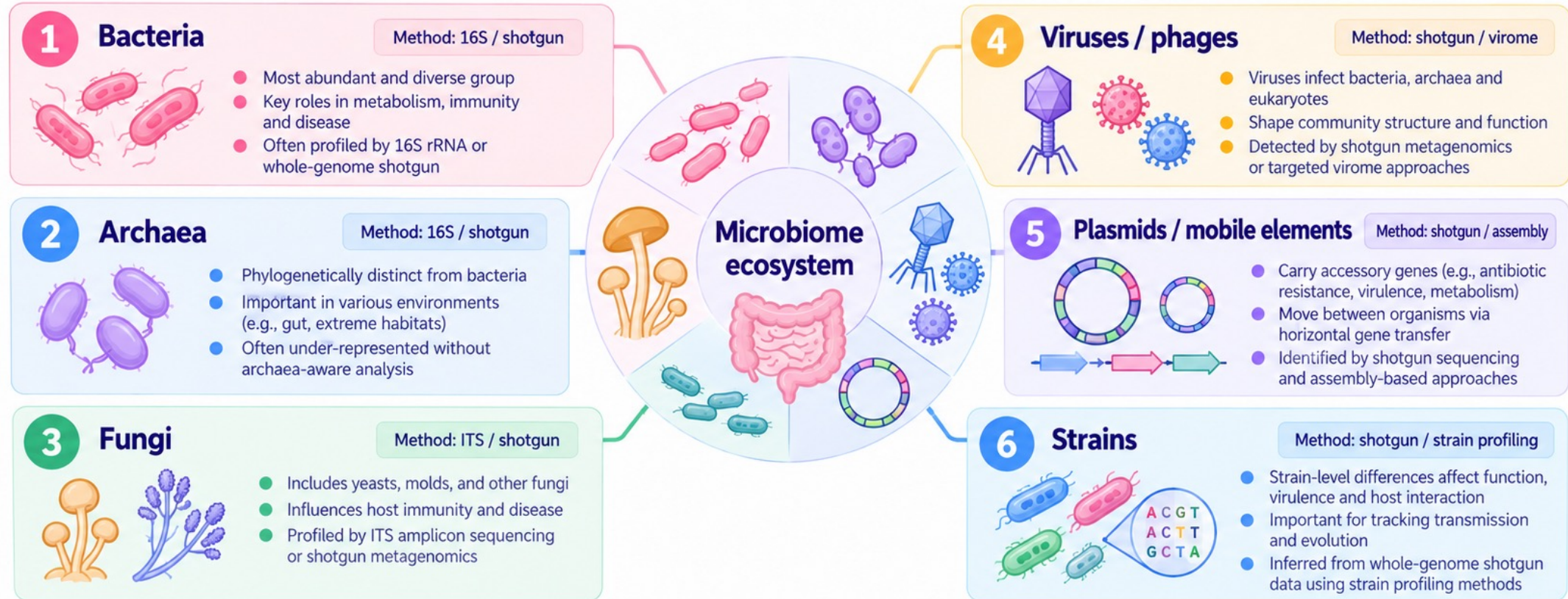
A metagenome-assembled genome is a computationally reconstructed genome hypothesis.



A MAG is a reconstructed genome hypothesis, **not automatically** an isolated organism.

The Microbiome Is More Than Bacteria

Different microbial and mobile genetic components require different measurement strategies.



What each layer can reveal



Community composition
Who is there?



Taxonomic breadth
How diverse?



Pathogenicity or symbiosis
What are they doing?



Horizontal gene transfer
How do genes move?



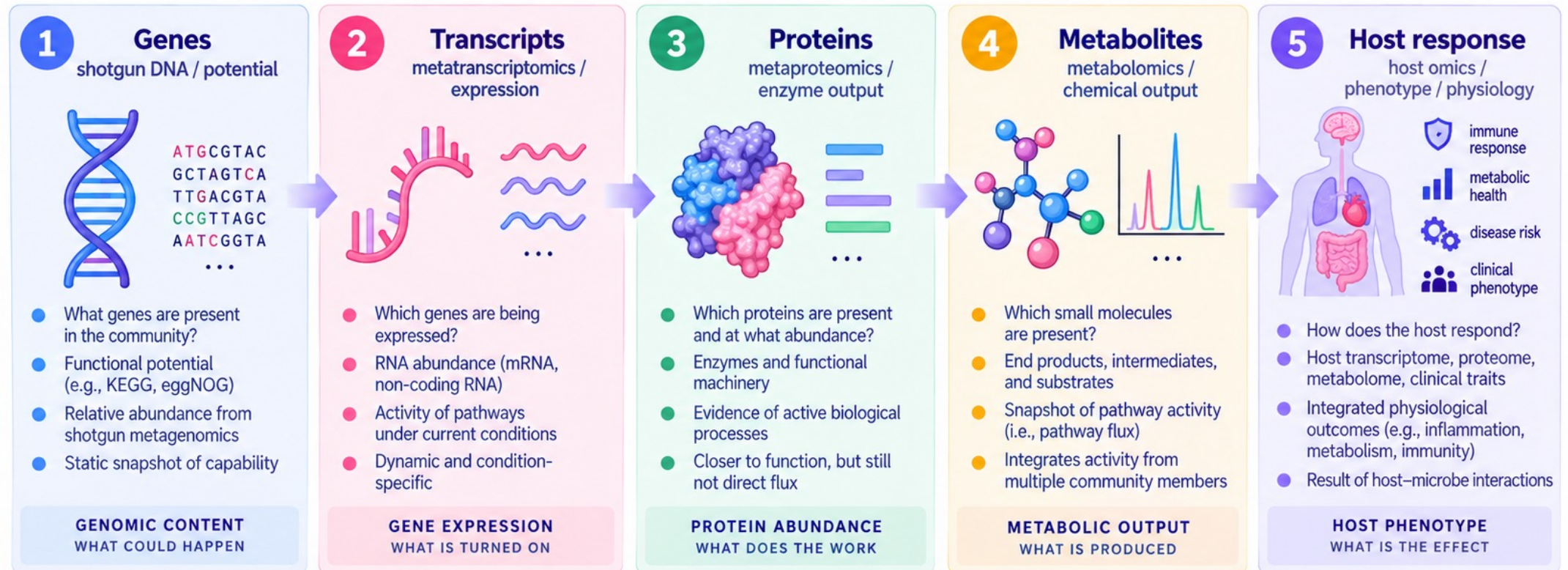
Microdiversity (strains)
Which variants matter?



Microbiome should **not** silently mean **only 16S-detectable bacteria**.

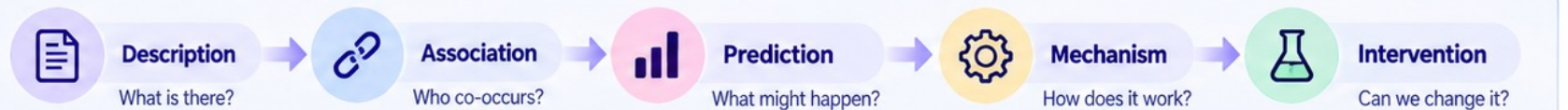
From Genes to Host Phenotype: Follow the Evidence Ladder

Encoded potential, expression, proteins, metabolites, and host response are distinct measurements.



How claims become stronger

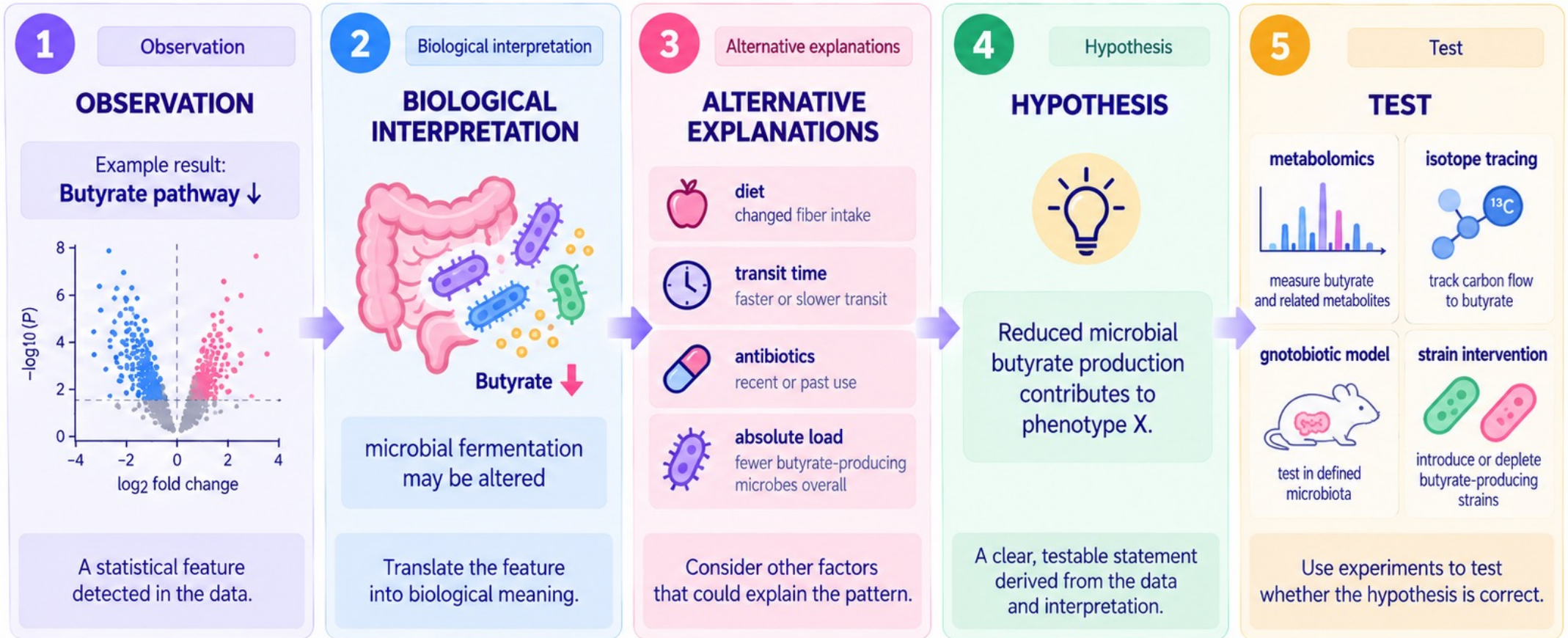
Each step provides more direct evidence, but different questions and uncertainties remain at each level.



A pathway encoded in DNA is **not the same** as pathway flux, metabolite concentration, or host effect.

A Significant Feature Is Not Yet a Scientific Question

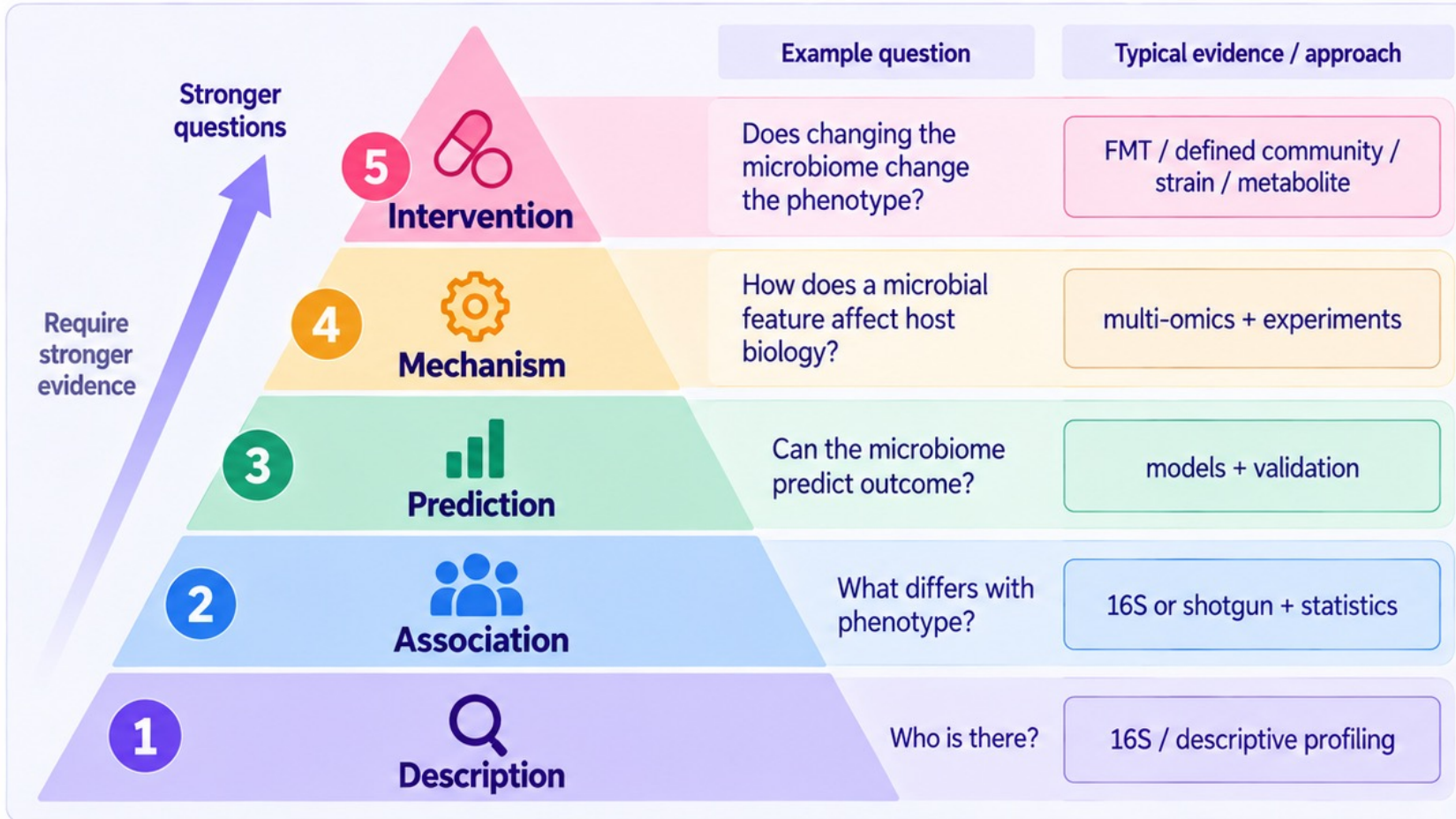
Bioinformatics identifies patterns; science begins when a pattern becomes a testable hypothesis.



A different feature is only the beginning; a scientific question requires **interpretation** and a **test**.

Five Levels of Microbiome Questions

As questions become stronger, the required evidence also becomes stronger.



Question upgrade = evidence upgrade

Moving up the pyramid shifts questions from “what is there?” to “does it cause a change?”, requiring increasingly rigorous data and validation.



Broader data

From community profiles to multi-omics and experiments



Stronger inference

From description to causation



Greater impact

Stronger evidence supports more actionable conclusions



Stronger microbiome questions demand **stronger evidence**.

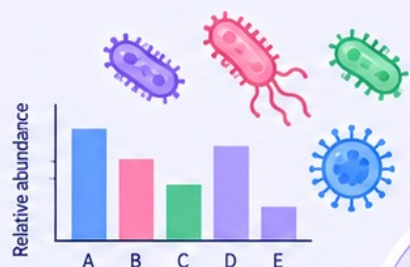
Multi-Omics Builds Stronger Biological Models

Integrated evidence can connect microbes, functions, metabolites, host pathways, and phenotype.

Taxa (microbiome)

Which microbes are present and how abundant are they?

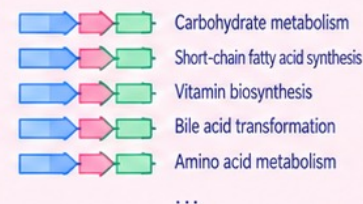
- 16S rRNA gene sequencing
- Shotgun metagenomics
- Taxonomic profiles
- Community structure



Genes / pathways

What functions can the microbes perform?

- Shotgun metagenomics
- Gene families (e.g., KEGG, COG)
- Pathway reconstruction
- Functional potential / activity



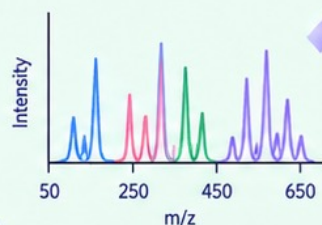
Phenotype

Host outcome

Metabolites

What molecules are produced or consumed?

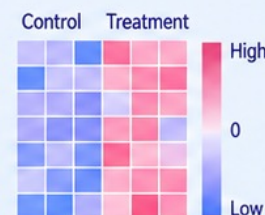
- Metabolomics (LC-MS, GC-MS)
- Microbial and host metabolites
- Pathway intermediates
- Functional readouts (e.g., SCFAs)



Host transcriptome

How does the host respond?

- Host RNA-seq
- Differential gene expression
- Pathway analysis (e.g., GO, KEGG)
- Host-microbe interactions
- Immune and metabolic pathways



Integrated mechanistic chain

Link multiple layers of evidence to build testable biological models.



Integrated correlation is not causality

Perturbation and validation are still required.

- Correlations may be indirect or confounded
- Shared environment and host factors
- Directionality cannot be inferred from multi-omics alone

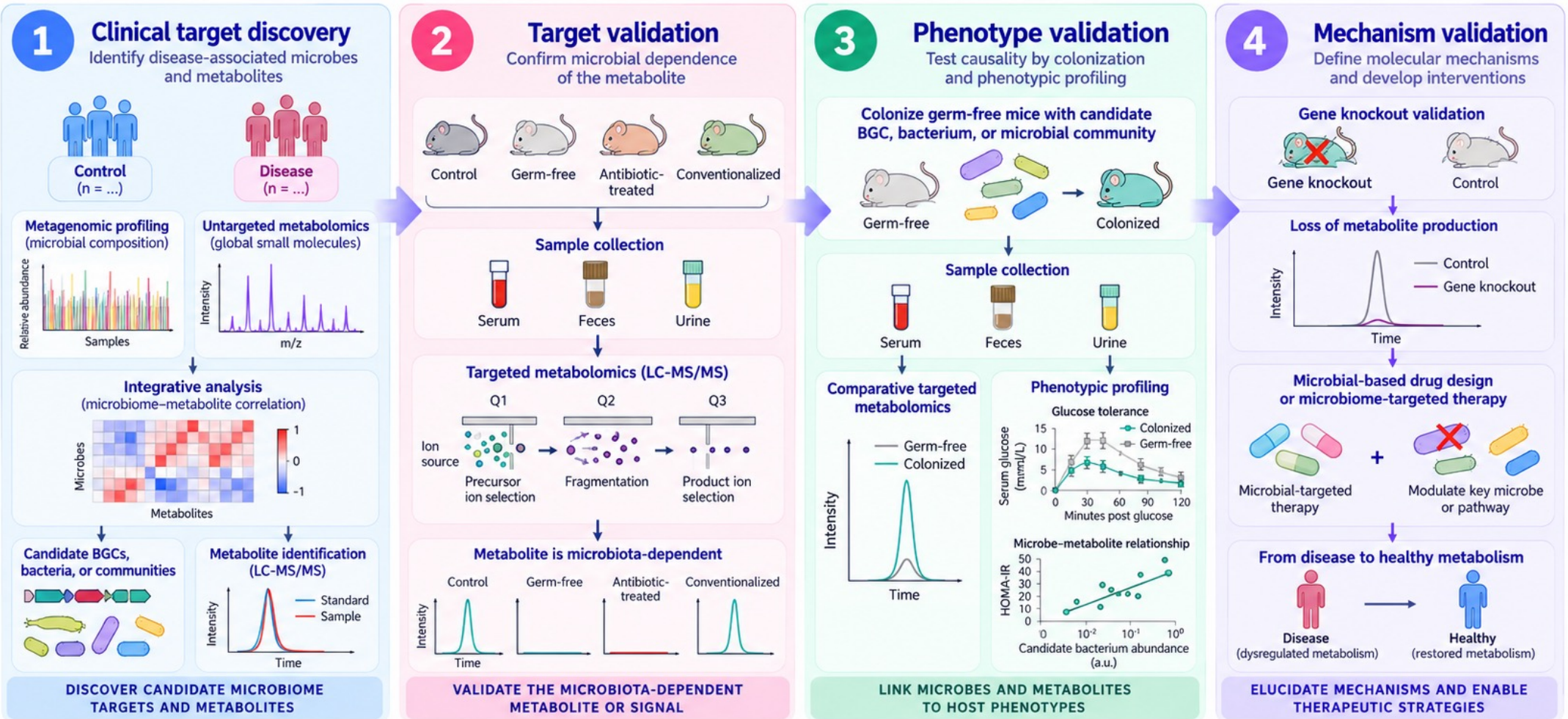
- Use perturbation (e.g., gnotobiotic models)
- Experimental validation
- Triangulate evidence across datasets



Multi-omics can strengthen inference, but **validation is still essential.**

Microbiome-Based Target Discovery and Validation

From clinical screening to target validation, phenotype testing, and mechanism-guided intervention.



Integrated microbiome, metabolomics, and validation workflows can turn clinical associations into **testable mechanisms** and **therapeutic strategies**.

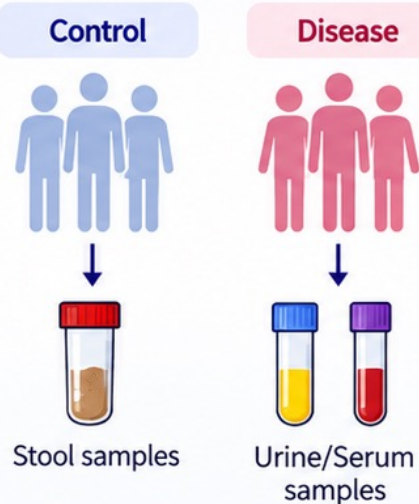
Adapted from Fan Y, Pedersen O. *Nat Rev Microbiol* (2020).
doi:10.1038/s41579-020-0433-9

1

Clinical cohort screening: identifying candidate microbial features

Metagenomics + Metabolomics for clinical phenotype association

1. Study design & sample collection

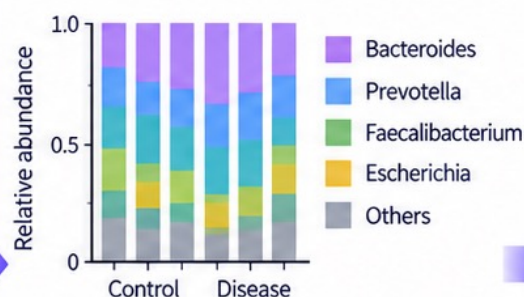


Data types

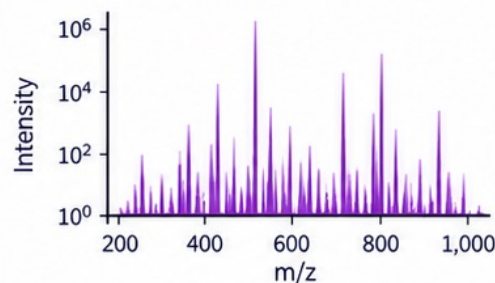
- Shotgun metagenomics
- 16S rRNA sequencing
- Untargeted metabolomics

2. Multi-omics profiling

Metagenomic profiling (taxa & function)

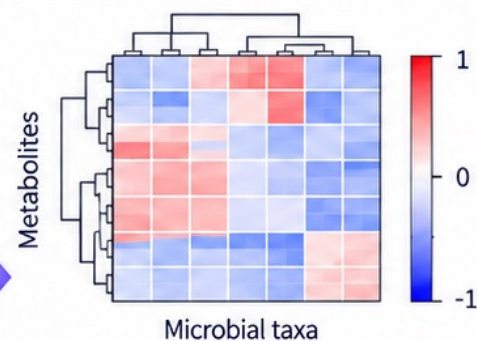


Untargeted metabolomic profiling

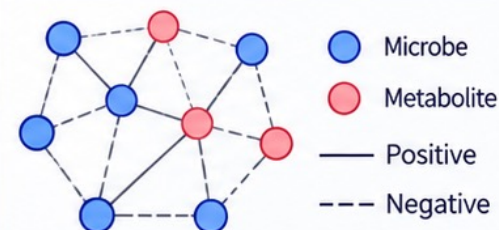


3. Integrative analysis

Microbiome-metabolome correlation

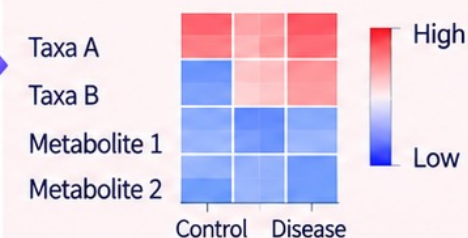
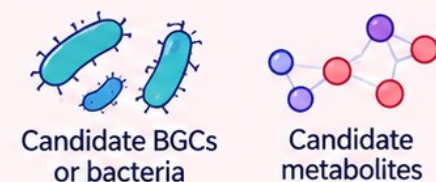


Network analysis



4. Candidate features

Disease-enriched features



Use in next step

- Select candidate strains/BGCs
- Targeted metabolomics validation
- Mechanistic studies



Take-home message: Integrated multi-omics analysis reveals microbiome features associated with disease and prioritizes candidates for validation.

2

Target validation: microbiota-dependent metabolite production

Germ-free mice and microbiota manipulation to validate microbial contribution

Experimental design



Conventional control



Germ-free



Antibiotic-treated

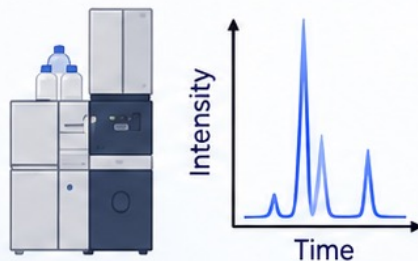


Colonized (defined microbiota)

Sample collection

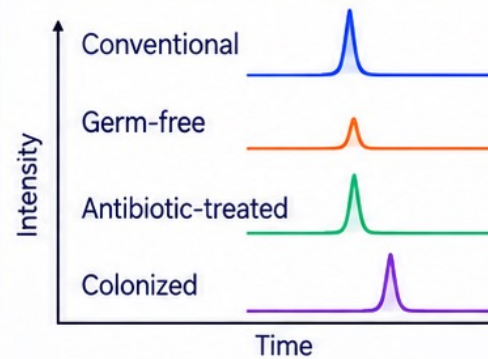


Targeted metabolomics (LC-MS / GC-MS)

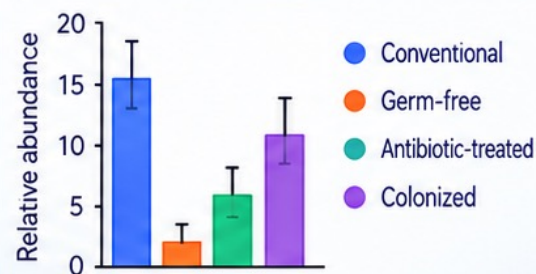


Data analysis

Metabolite comparison



Quantification



Key findings & interpretation

- Metabolite present in conventional mice
- Dramatically reduced in germ-free mice
- Restored after colonization
- Indicates microbiota-dependent production

What can we conclude?

- ✓ Microbes are necessary for metabolite production
- ✓ Specific taxa/communities can restore the metabolite
- ✓ Provides causal evidence for microbial contribution



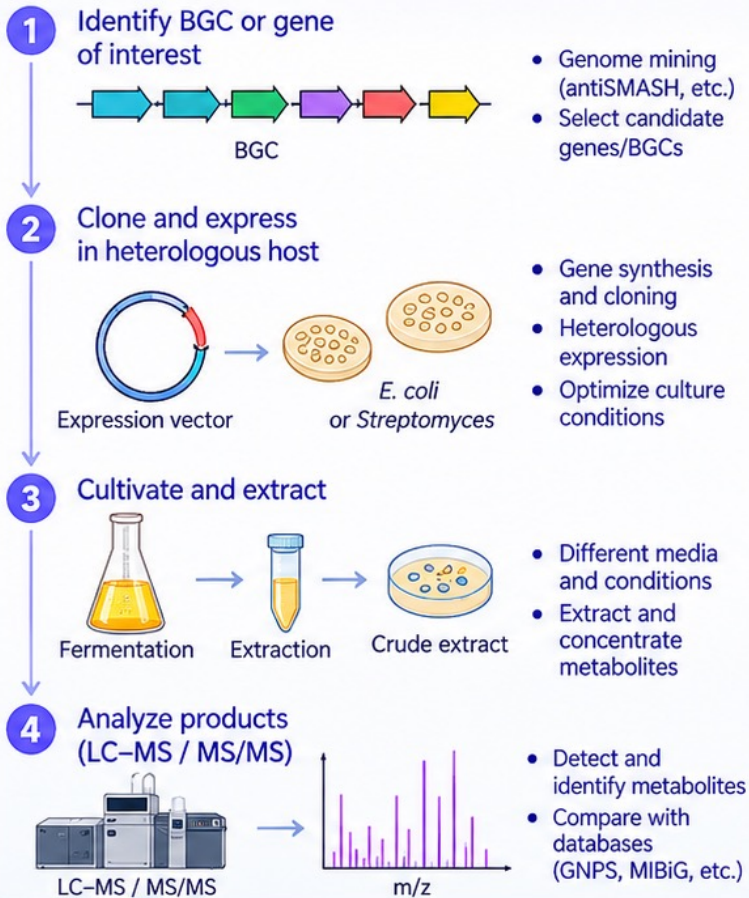
Take-home message: Germ-free and colonization experiments validate whether a metabolite is produced by the microbiota.

3

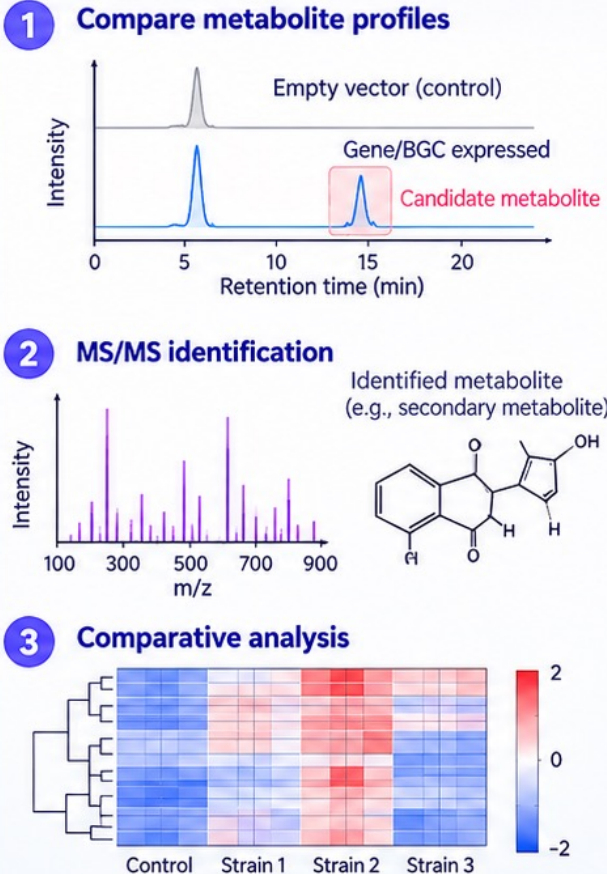
Functional characterization: expressing microbial genes and BGCs

From gene to molecule: heterologous expression and functional validation

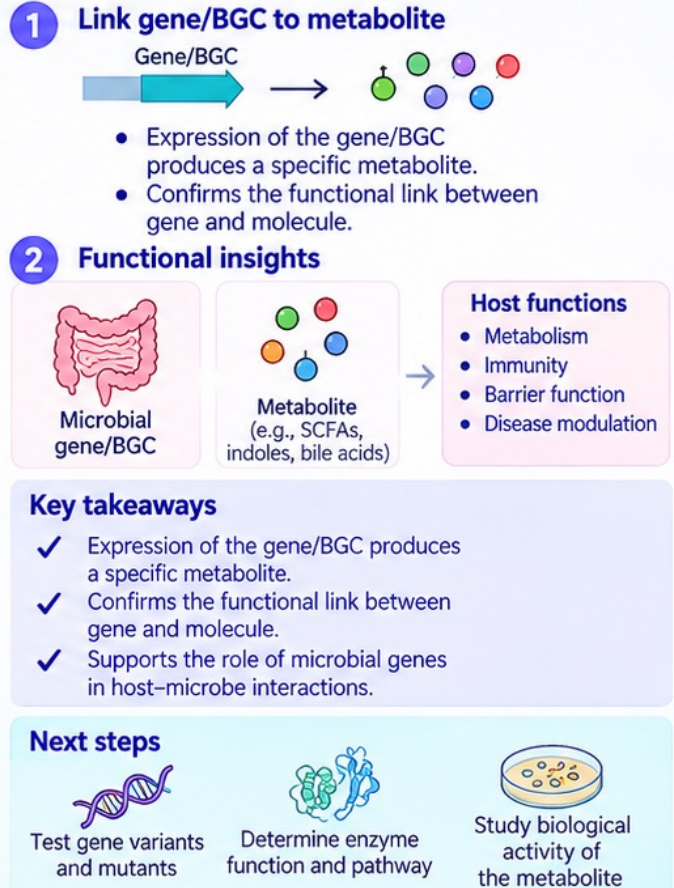
Workflow overview



Data analysis



Biological interpretation



Take-home message: Heterologous expression links microbial genes/BGCs to their metabolic products, providing functional evidence.

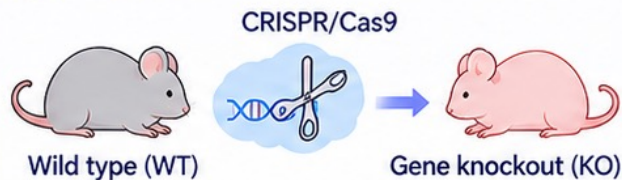
4

Mechanistic validation: gene knockout and microbiome-based therapy

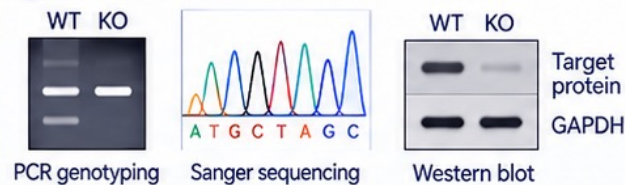
From mechanism to intervention: functional validation and translational potential

Genetic validation

1 Target gene deletion



2 Confirm knockout

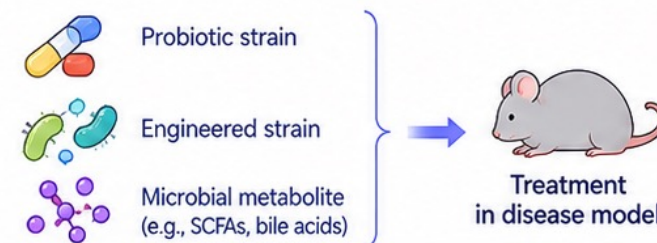


3 Functional characterization

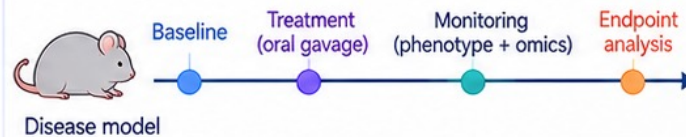


Therapeutic validation

1 Microbiome-based intervention



2 Experimental workflow

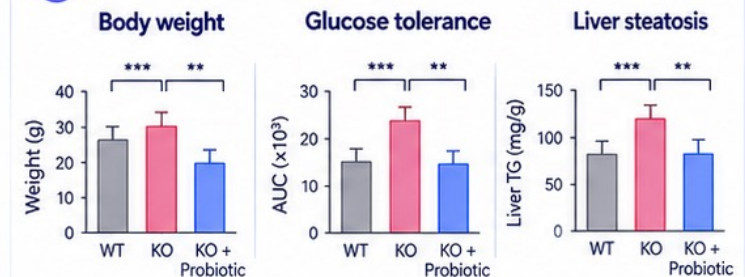


3 Outcome evaluation

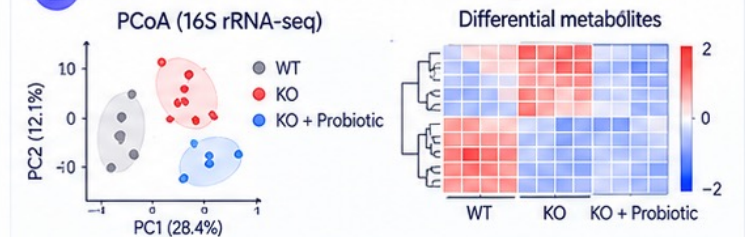


Data analysis & interpretation

1 Phenotypic outcomes



2 Microbiome and metabolome analysis



3 Key insights

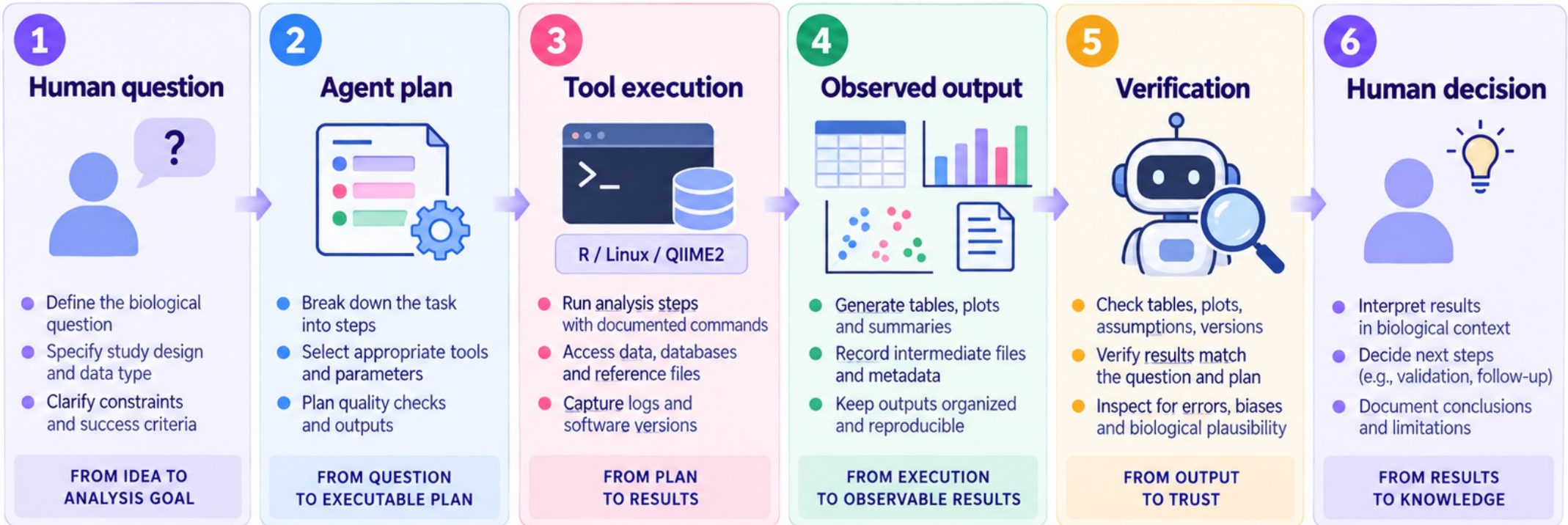
- ✓ Loss of microbial metabolite production contributes to disease phenotype.
- ✓ Microbiome-based therapy can restore key metabolites and improve metabolic health.
- ✓ Provides mechanistic evidence and translational potential for microbiome-targeted therapy.



Take-home message: Functional validation and intervention studies establish causal mechanisms and guide microbiome-based therapies.

AI and Agents Can Accelerate the Workflow

Automation is useful only when microbiome decisions remain transparent and inspectable.



Mandatory checkpoints

Every step should be transparent, reproducible, and inspectable.



inputs

data files,
parameters



IDs

sample IDs,
feature IDs



controls

negative/positive
controls, batches



method

software, version,
reference database



statistics

tests, correction,
effect sizes



biology

biological
plausibility



outputs

tables, figures,
reports



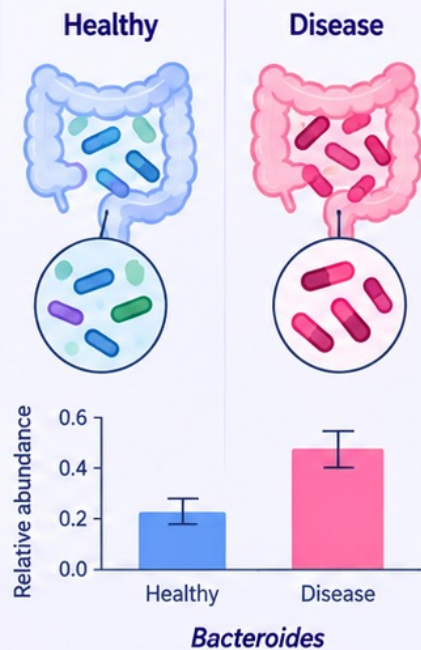
AI accelerates execution; it **does not replace** provenance, source checking, or scientific judgment.

Use AI to Generate Hypotheses, Not to Replace Thinking

AI can suggest explanations, but students must evaluate evidence, alternatives, and falsification tests.

Observed result

Bacteroides abundance is associated with disease.



AI-generated hypotheses

AI can propose multiple, plausible mechanisms for the observed association.

H1 Diet-mediated

Disease is associated with dietary factors that increase *Bacteroides* abundance (e.g., high fiber or fat intake).

H2 Bile-acid-mediated

Altered bile acid composition in disease favors *Bacteroides* growth and colonization in the gut.

H3 Inflammatory niche-mediated

Disease-related inflammation creates a gut environment that supports *Bacteroides* expansion.

Supporting evidence

Look for data that support each hypothesis.

- Human diet-microbiome studies
- Higher *Bacteroides* with high-fiber diets
- Consistent findings across cohorts

- Altered bile acids reported in disease
- *Bacteroides* bile-acid tolerance genes
- Animal or *in vitro* support

- Inflammation markers correlate with *Bacteroides*
- Mouse colitis models show increased abundance
- Mechanistic studies on niche adaptation

Alternative explanation

Consider other reasons for the association.

- Confounded by overall diet or lifestyle
- Association not causal
- Co-varying microbes drive the signal

- Bile acid changes are consequence, not cause
- Other taxa respond similarly
- Host genetics or medications explain trend

- Inflammation results from disease, not microbes
- Other pathobionts are the primary drivers
- Reverse causality cannot be excluded

Falsification experiment

Design a test that would disprove the hypothesis.

- Control diet in a longitudinal study
- Test multiple diets in gnotobiotic mice
- If diet is controlled and *Bacteroides* still differs, H1 is weakened

- Manipulate bile acids in mice (e.g., inhibitors)
- Test *Bacteroides* growth in defined media $\pm$ bile acids
- If abundance does not change with bile acids, H2 is falsified

- Reduce inflammation in a disease model
- Test if *Bacteroides* remains high after anti-inflammatory treatment
- If abundance normalizes with reduced inflammation, H3 is weakened



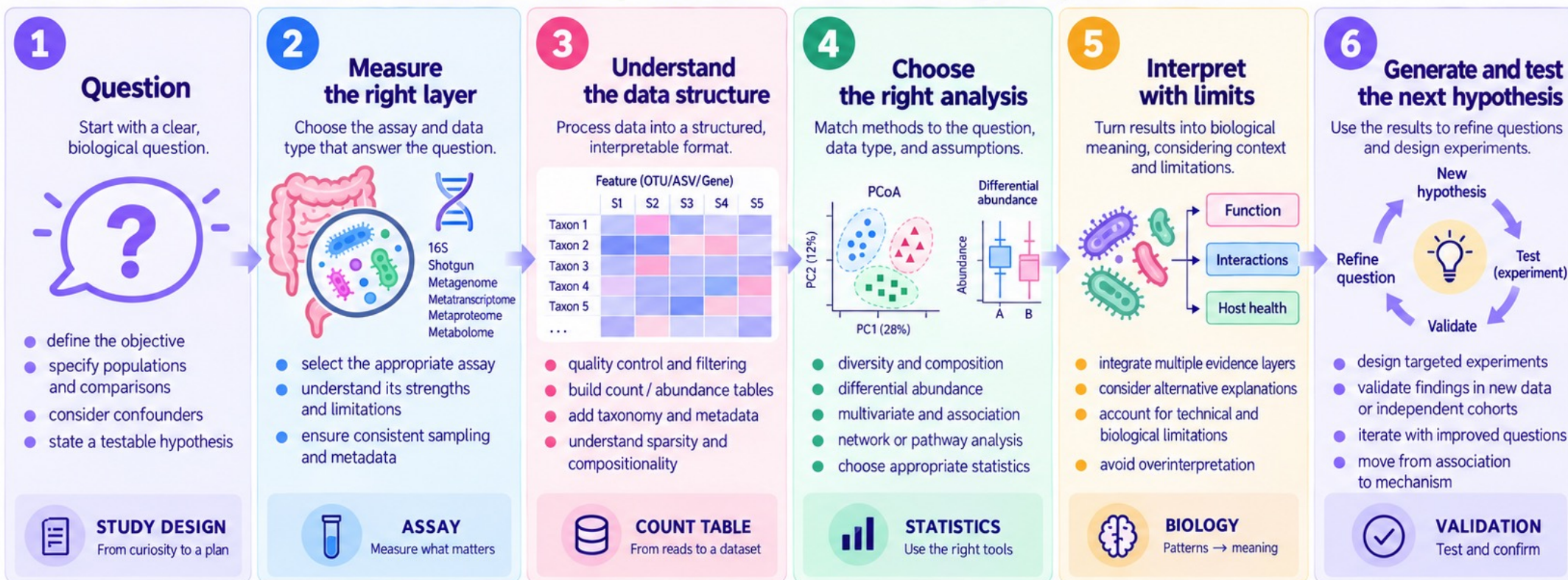
Use AI as a hypothesis generator and critical-thinking partner, **not as an answer machine.**

Question First. Evidence Always.

Microbiome bioinformatics turns community data into testable biological questions.

Data • Microbes • Ideas • Healthier Lives

BIOINFORMATICS FOR A HEALTHIER PLANET



Week 6 take-home

Key lessons to carry forward into your own research.



1. assay shapes the data

The biological layer and method determine what you can see.



2. plots are not conclusions

Visual patterns need statistical support and biological context.



3. patterns should lead to hypotheses and tests

Use insights to design the next round of experiments or data analyses.

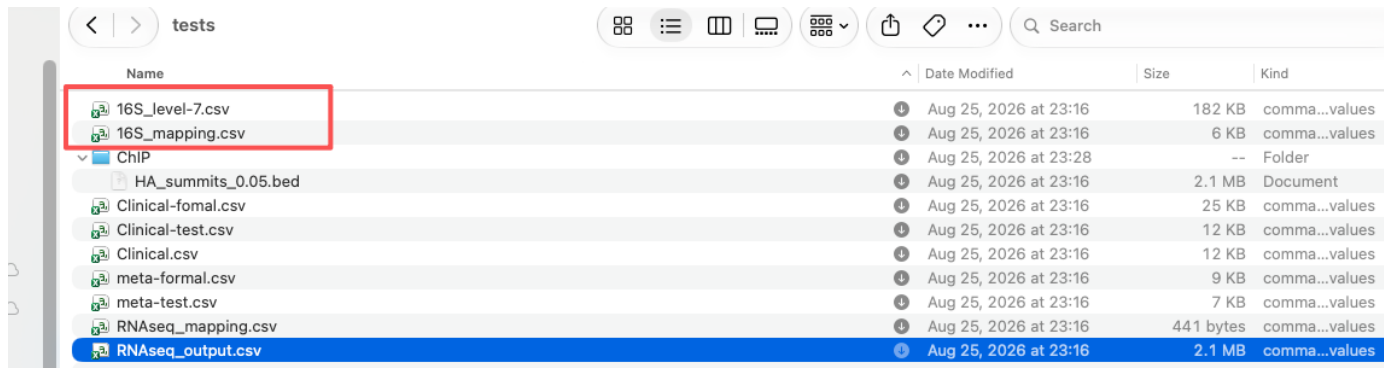


Microbiome bioinformatics is the process of turning data into **testable biological questions**.

Homework for week 5/6, **Submission deadline 23:59 on Sunday Night**

Homework : Use 16S data files in EMP tests folder and EMP-web to run analysis.

1. You need to complete whole analysis procedures and submit the final result via the EMP-web system.
2. Based on the parameter and analysis results, please generate a scientific hypothesis based on your analysis results. You need to indicate what kind of parameters you use to support your hypothesis



Name	Date Modified	Size	Kind
16S_level-7.csv	Aug 25, 2026 at 23:16	182 KB	comma...values
16S_mapping.csv	Aug 25, 2026 at 23:16	6 KB	comma...values
ChIP	Aug 25, 2026 at 23:28	--	Folder
HA_summits_0.05.bed	Aug 25, 2026 at 23:16	2.1 MB	Document
Clinical-fomal.csv	Aug 25, 2026 at 23:16	25 KB	comma...values
Clinical-test.csv	Aug 25, 2026 at 23:16	12 KB	comma...values
Clinical.csv	Aug 25, 2026 at 23:16	12 KB	comma...values
meta-fomal.csv	Aug 25, 2026 at 23:16	9 KB	comma...values
meta-test.csv	Aug 25, 2026 at 23:16	7 KB	comma...values
RNAseq_mapping.csv	Aug 25, 2026 at 23:16	441 bytes	comma...values
RNAseq_output.csv	Aug 25, 2026 at 23:16	2.1 MB	comma...values

We will have quiz next week.

Do not forget to review the class material.

5 multiple choice questions

1 min of each question for total 5 mins